Truck Operations Manual

Fire Chief Forward

The PFD Truck Company Operations Manual is a guideline to provide success for our truck company crews. This manual provides guidance allowing PFD to operate in an aggressive, yet calculated, supporting actions within fire attack operations. To establish the "Pearland Way" by setting clear expectations and understanding for "yet calculated" is that we expect all members, and we will hold each other to the standard of aggressively performing all tasks, duties, and assignments to the full extent possible within each members physical abilities, protective equipment, and training. This allows us to achieve the highest level of success within our goals of preservation of life, scene stabilization, and property conservation.

Utilizing the philosophies within this manual, PFD Truck companies will function in a professional mindset and routinely display their technical excellence with the goal of mitigating any obstacle that stands in our way. Assignment to a Truck Company is an honor preceded by a rich tradition within the fire service. As a “Truckie,” you will earn the honor of representing “the shamrock” within PFD. Together, we will set the standard of “Truck Work” that other organizations can emulate, but never duplicate. You are a select few that are honored to serve on a Pearland Truck Company. Be proud of that role and embrace it.

In 1845 the Great Potato Famine struck Ireland and led to mass immigration of the Irish into America. During the famine years, nearly one million Irish arrived in America. As you may know, the Irish settlers were not exactly welcomed to the US. Many of the settlers were mocked and faced hard times in America. During this period there were a limited number of unskilled jobs available, and many Americans were worried that the Irish immigrants would be willing to work for lower wages and undercut them for jobs. This led to an increase in anti-Irish sentiment in cities like New York and Boston. Many employers posted “No Irish Need Apply” signs to keep these unwanted guests from taking American jobs. The only jobs these Irish immigrants could get were the dirty and dangerous civil service jobs that no one else wanted. As a result, many Irish immigrants became firefighters and police officers.

Irish American firefighters began affixing images of the shamrock to their apparatus and their person not only as a display of Irish-American pride, but also as an inconspicuous message to their fellow Irishmen advertising that the fire service is a place that won't discriminate against them. The same stands at PFD. We will not discriminate and welcome all from every walk of life who are able and passionate in answering the call to serve our community. Today, by tradition, most truck companies have a shamrock somewhere in their logo, on their apparatus, or on their helmet, as well, to channel the "luck of the Irish".

Carry this assignment with pride, represent your brothers and sisters, and protect them using the tools and skills provided in our PFD Truck Operations Manual. Good luck and be safe.

J. Taylor
Chief of Department

J. Taylor - Chief of the Department

C. Birt - Assistant Chief of Operations

J. Johnson - Assistant Chief of Administration

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Section I: Truck Company Overview

Our department's success on the fireground hinges on the skills and abilities of all personnel operating; this manual focuses on those specific to members assigned to a truck company.

The role of a truck company at the scene of a building fire is to ensure the rapid search and removal of fire victims from the hazard zones while supporting the extinguishment efforts of the fire. The truck company's tasks are numerous, and the survivability of civilians and other firefighters is increased when the truck company executes assigned duties effectively, coordinated, and smart.

The truck company generally works in close proximity to, and above the fire, often without the protection of a hose line. Supporting the engine company is key, therefore, roles are directed towards locating the fire, containing it if necessary, and searching for occupants. Other members remain outside to throw ladders, ventilate, and operate the aerial, if needed. Many of these functions need to be conducted in a coordinated effort with fire attack to ensure civilian and firefighter safety, as well as property conservation.

Some of the tasks outlined in this manual are the classic "high risk, low frequency" events. These events can be extremely dangerous and will present enormous life safety factors, regarding both civilians and firefighters. Decisions will need to be made immediately, and it is imperative that truck companies performing the duties outlined in this manual are well trained, efficient, and capable of performing these duties without fail.

PFD has an aggressive truck culture that allows its Truck Company to perform their assigned duties to the highest level they deem intelligent, worthwhile, and within their capabilities as members and as a company. Following our risk management doctrine, all Truck Company members will remember and understand that "You can be ordered to the roof, but you cannot be ordered on the roof." While this does not only apply to the roof it allows each member to use real-time situational awareness and specific location authority to determine if an order can be accomplished smartly, successfully, and purposefully.

Section II: Truck Company Philosophy

Pearland truck companies will operate with the philosophy that aggressive smart supporting actions assisting fire attack operations will allow the highest probability of preservation of life, scene stabilization, and property conservation. Aggressive and solid firefighting relies heavily on the truck company's ability to search for and locate the fire quickly. This will also aid in rapid fire suppression and rescue of trapped victims from an increasingly non-survivable environment.

Section III: Definitions and Terms

Aerial Ladder: A power-operated ladder that allows firefighters to easily climb and descend between the turn table and the tip of the ladder.

Collapse Zone: The area around a building, at least one and one-half times the height of the building involved, that should be considered as a danger area for potential collapse when heavily involved in fire or the building has been significantly damaged.

Conventional Wood Floor/Roof Construction: Construction methods using dimensional lumber for structural members that depend on mass for strength.

Coordinated Ventilation: The controlled removal of smoke, heat, and gases and the replacement with fresh air in concert with deployment and application of a properly functioning hose line on the seat of the fire.

Elevated Master Stream: A fire stream more than 350 GPM that is deployed from the tip of an aerial device.

Flying Standpipe: The use of the pre-piped master stream or laying a larger line (2.5-inch or 3-inch) up the aerial to serve as a standpipe to deploy a handline.

Forcible Entry: The act of gaining access into a structure or occupancy via door, window, or wall by use of force.

Green Building: A structure or occupancy designed to be environmentally responsible and resource efficient throughout the building's life. A green building makes efficient use of resources, such as water and energy, and, in doing so, lessens the building's impact on the environment.

Ground Ladder: A portable ladder designed to rest on the ground to provide firefighters with means of access and egress above and below ground. The most common include straight ladders, extension ladders, and folding ladders.

Stabilizers/ Outriggers: These are deployed to stabilize the truck when the aerial device is in operation.
Definitions:

Search and Rescue: a coordinated, systematic process used to locate, protect, and safely remove occupants and victims from an environment that presents to them an imminent risk of death or serious bodily harm, and move them to a place of safety.

Secondary Search: must be completed as soon as possible after the completion of a primary search. This search is a thorough and complete inspection of the area searched during the primary search.

Halyard: natural fiber, non-weight rated rope used to raise and lower fly sections. Halyard should not be tied in the case of a victim rescue when a life is threatened.

Healing: the act of securing a ladder from the base on the ground while another is using the ladder.

Horizontal Ventilation: Any technique by which heat, smoke, and other products of combustion are channeled horizontally out of a structure by way of existing or created horizontal openings such as windows, doors, or other holes in walls.

Inspection Cut: A small cut used to determine conditions under the roof such as: nature of the smoke (i.e., color, volume, temperature, pushing under pressure); volume of fire; the location of fire; direction of fire travel; the type, size, roof thickness, rafter spacing, and run of the structural elements; and the fire's extension.

Ladder Bridging: Using a ground ladder to span distance resembling a bridge. This can be used to span buildings, trenches, etc.

Ladder Splicing: Using a straight ladder in conjunction with an extension ladder when the extension ladder is not long enough to reach a fire victim. This should be done with extreme caution and only when lives are in imminent danger.

Ladder Tower: An aerial device that contains a full-size climbing (tubular truss beam construction) ladder that also has a basket/bucket for firefighting operations attached to the tip.

Lightweight Roof Construction: This lightweight truss construction does not derive strength from mass, instead strength is obtained from multiple members that are in compression and tension. The strength of the individual structural member is dependent on the total sum of the other members; therefore, if one member fails, others may fail.

LIP: Acronym for Life safety, Incident stabilization, and Property conservation, which serve as the primary incident priorities.

Mid-mount Aerial: An aerial ladder that has the turntable mounting on the chassis in the middle of the vehicle, usually behind the cab.

Negative Pressure Ventilation: Using smoke ejectors to develop artificial circulation and to pull smoke out of a structure. Smoke ejectors are placed in windows, doors, or roof vent holes to pull the smoke, heat, and gases from inside the building and eject them to the exterior.

Parapet: A wall-like barrier at the edge of a roof, terrace, balcony, or other structure.

Positive Pressure Ventilation: The use of fans to pressurize a structure or compartment in conjunction with the opening and closing of doors and windows to effectively create a flow path to remove smoke from a structure.
Primary Search: a rapid search of all involved and exposed areas affected by the incident.

Overhaul: The practice of opening up walls, ceilings, floors, or any void area where fire may have entered during the firefight. These areas must be aggressively opened up, inspected, and extinguished to avoid rekindle.

Quint: Fire apparatus that is equipped with a fire pump, water tank, ground ladders, fire hose, and an aerial device.

KNOX Key Entry System: A secure device with a lock operable only by a fire department master key and containing building entry keys and other keys that may be required for access in an emergency.

Rear-mount Aerial: An aerial ladder that has the turntable mounted at the rear of the chassis.

Roof hooks: round steel bar bent into a hook shape. Hooks fold using a push (towards foot-end) and rotate. Hooks can be slid over roof peaks or windowsills to assist in securing the ladder.

Roof Pitch: A numerical measure whereby the rise (vertical distance) and the run (horizontal distance) determine the steepness of the roof.

Salvage: The protection of buildings and their contents from unnecessary damage due to water, smoke, heat, and other products of the fire.

Scrub Area: The area of the building that the aerial tip or elevated platform can effectively reach.

Short Jacking: A method by which the ground support devices, stabilizers, or outriggers are not deployed the full distance on the non-working side because of an obstruction or topography.

Tower Ladder: An aerial device with a basket (box and beam construction) attached to the end of a boom that also contains a small ladder attached to the boom that can be used for escape.

Tractor Drawn Aerial: A tractor-trailer aerial apparatus, also known as a tiller truck, that is equipped with steerable rear wheels on the trailer.

Turntable: A rotational structural component of the aerial device allowing continuous rotation on the horizontal plane. Utility Control: The act of shutting off the gas, electricity, and/or water to the structure or occupancy.

Vent for Search: Situation where firefighters create openings, or break windows, to gain access from an exterior position to carry out a primary search in a high-risk area of the structure.

Vent for Extinguishment: Improving interior conditions for firefighters by reducing heat levels and improving visibility.

Ventilate Enter Search (VES): Situation where firefighters create openings, or break windows, to gain access from an exterior position to carry out a primary search in a high-risk area of the structure. This will be conducted with the use of ladders for areas above ground level.

Vertical Ventilation: Ventilating at the highest point of a building through existing or created openings and channeling the contaminated atmosphere vertically within the structure and out the top.

Section IV: Truck Company Tools and Equipment

Minimum Ladder Truck Equipment List and set up.

All PFD Ladders, Tower Ladders and other apparatus will always be set up for offensive operations. All deck guns, monitors and ladder pipes will be equipped with smooth bore nozzles for immediate needs. Waterways will be pinned in Rescue Mode. Changes to other nozzles can be made if needed for special operations. The Tower with two ladder pipes will remain one of smooth bore and one Fog.

This standard will apply to all Aerial Apparatus within the Pearland Fire Department, excluding previously equipped Quints.

- Minimum of 160 ft of ground ladders
  - Two folding ladders
  - Three straight ladders
  - Three extension ladders
- One handheld radio and strap each riding position.
- Four box flashlights
- Four right angle flashlights
- One SCBA for each riding position
- One RIT Pak II
- Six spare SCBA cylinder for each riding position
- Four ladder belts
- One traffic vest for each riding position
- One ballistic vest for each riding position
- Two TICS
  - Two TIC Chargers
  - Two Spare TIC batteries
- 5 gas meter
  - Gas meter mount
  - Gas meter charger
- Binoculars
- Cooler
- Two wheel chocks
- One AED or cardiac monitor
- One medical bag
- Eight DOT fluorescent orange traffic cones
- Five illuminated warning devices, such as highway flares
- Six salvage covers.
  - Minimum 12 ft x 18 ft
- Three disposable tarps
- Two Dry Chem extinguisher
  - 80-B:C
- Two water extinguishers
- One CO2 Extinguisher
- Two 8 lb lock slot 8 flathead axes
- Two 8 lb “PIG” axes
- Nine hooks
  - Two 6 ft NY hook
  - One 6 ft Boston rake
  - Two 8 ft NY hook
  - One 8 ft Colorado hook
  - Two 10 ft NY hook
  - One 12 ft NY hook
- Two 3-4 ft dry wall hooks with D handles
- Two 10 lb sledgehammers
- Three Pro-Bar Halligans
- 51” pinch point pry bar
- Two scoop shovels

- Two spanner wrenches
- Two large spanner wrenches
- One 36” Bolt cutter
- Two STIHL 462 Chainsaws w/ RDR chains
    - With associated tools and fuel
    - Spare RDR chain
- Two Husqvarna K12 970
    - One with a diamond blade and one with a demo carbide toothed blade.
    - Spare blades
- Milwaukee 9” battery saw w/ diamond blade
- Two plug in scene lights
- Two battery powered scene lights
- Two 200 ft Life safety rope bags
- One 150 ft Life safety rope bag
- 300 ft combined utility rope
    - Minimum 5000 lb breaking strength.
- Rope rescue hardware bag
    - Fourteen CMC Large Carabiner
    - One CMC X-Large Carabine
    - Two CMC Ascender
    - Three CMC Rope Minding Single Pulley
    - Two CMC Rope Minding Double Pulley
    - One CMC Rescue Swivel
    - One set of Locking Trilink and Bull Ring
    - CMC Rescue 8
    - CMC Brake Bar Rack
    - CMC Anchor Plate
    - Plastic Edge Protector
- Rope software bag
    - Seven 8mm Pre-cut Prusik/Accessory Cord
    - Two Webbing Pre-cut Stokes lashing YELLOW
    - Two Webbing Pre-cut Stokes lashing BLUE
    - Two Webbing Pre-cut Stokes lashing RED
    - Three CMC 6' Anchor Strap
    - Two CMC 4' Pick-off Strap
- Stokes Basket
    - With CMC commercial rescue bridal
- Rescue half back
- Rescue Sked
- Four class III harness
- Mechanic wrench and socket set
- One assorted tool bag
    - 8-Pack Assorted Plier Set
    - VISE-GRIP 2-Pack Locking Plier Set
    - Wire Strippers
    - 25-key Folding Hex Key Set (Allen wrenches and Torx set)
    - Utility Knife with On Tool Blade Storage
    - Tape measures 25-ft
    - 110v to 125-Volt Analog Dual Range Non-Contact Voltage Detector
    - Three 7-in Zippered Tool Bag (for organization)
    - 16-oz Smoothed Face Steel Hammer
    - 500-Pack zip ties
    - 25-Pack Multi Wire Connectors (wire nuts)
    - Heavy Duty Rubberized Duct Tape
    - Electrical Tape
    - Straight Cut Tin Snips
    - Mini 10-in Medium Cut Hack Saw
    - Spyder Diamond Edge 2-Pack Diamond 4.5-in Cut-off Wheel
- Roof patch tool bag
    - Heavy Duty Manual Staple Gun
        i. Box of staples

  - One box of Galvanized Steel Roofing Nails
  - One box of 2-in Galvanized Steel Common Nails
  - One box of #8 x 2-1/2-in Wood To Wood Deck Screws
  - One roll of 10-ft x 25-ft 6-mil Construction Film
  - 22-oz Smooth Face Steel Hammer
- Two push brooms
- Three sets of Struts
  - assorted tips
  - Four base plates
  - Six sets of ratchet straps
  - Two shoring channel base for 4”x4”
  - Two chains 10mm x 6.1m w/ hook
  - Two chain binders
- Four Step Chocks
- Eight 4x4 cribbing
- Sixteen 2X4 cribbing
- Eight 2x4 wedges
- Two PPV fans
- Glass master
- Hydraulic Extrication tools
  - Cutter
  - Spreader
  - Ram
    i. With support
  - Hydraulic pump with two hoses
  - Battery Mini Cutter
- Two search rope bags
- Power tools
  - Drill
    i. With drill and drive bit set
  - ¼” Impact Drill and ½” Impact Drill
    i. Impact socket set
  - Reciprocating saw
    i. Saw blades
  - Angle grinder
    i. With cut off disk
- Big Easy
  - Elevator Key Roll
- K- tool Kit
- S &D Rex Tool
- Costal Fire Training Complete Respectful Entry Kit
- Hydra Ram
- Rescue air chisel
- One PFD per riding position
- One Throw bag
- Five set lifting air bags
  - Control unit
  - Regulator
  - Multicolored hoses
  - Four Inline relief valves
- One MultiForce lifting bag set w/ remote placement handle

Section V: Truck Company Arrival Assignments

Fire Alarms

- 1st Arriving Truck
    - Rapid 360 walk around of structure when possible and applicable
    - Ensure the structure has been evacuated/ start evacuation of the structure.
    - Assist Engine with investigation.

Single Residential Structures

- See deployment model.

Multi-residential Occupancies

- See deployment model.

Commercial Occupancies.

- See deployment model.

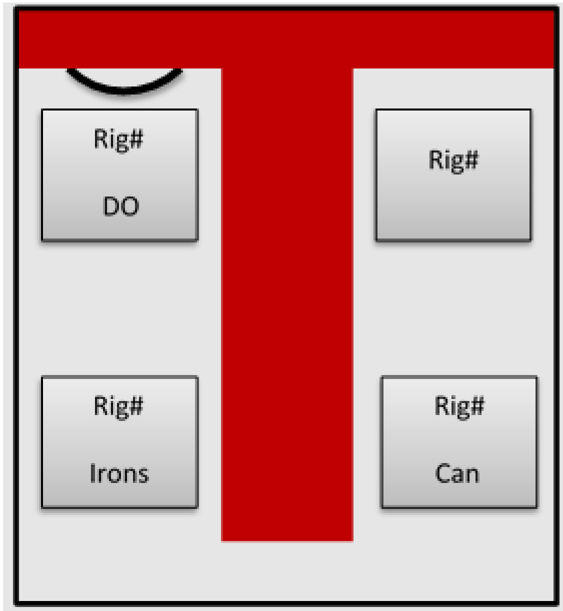
Technical Rescue

- 1st Arriving Truck
    - 360 of the scene if applicable
    - Ensure control zones are established and communicated.
    - Establish air monitoring as needed.
    - Establish rescue plan.
    - Establish lowering/hauling systems depending on needs.
    - Call for additional resources or any needed assistance.
    - Initiate rescue.
- 2nd Arriving Truck
    - Monitor air supplies if needed to effect rescue.
    - Establish back up rescue team (RIT)
    - Establish plan for secondary system.

Section VI: Truck Company Riding and Tool Assignment

Riding Assignments

Riding Location and Radio ID



Tool Assignments

Riding Assignment	Alarms	Single Residential Fires	Mulit-Residential/ Commercial Fires	Extrications	Technical Rescues
Officer Radio ID= Rig #	Radio 2 flashlights SCBA & TIC Forcible entry/ exit tool	Radio 2 flashlights SCBA & TIC Forcible entry/ exit tool	Radio 2 flashlights SCBA & TIC Forcible entry/ exit tool	Radio	Radio Rescuer Harness
Can Radio ID= Rig# Can	Radio 2 flashlights SCBA Hook Extinguisher *Search Rope	Radio 2 flashlights SCBA Can Hook Ladder (roof/ egress)	Radio 2 flashlights SCBA Can Hook Ladder (roof/ egress) *Search Rope	Radio Rescue Tools	Radio Rescuer Harness Rope Bags

Irons Radio ID= Rig# Irons	Radio 2 flashlights SCBA & TIC Irons Respectful entry kit	Radio 2 flashlights SCBA & TIC Irons Saw (ventilation)	Radio 2 flashlights SCBA & TIC Irons Saw (ventilation) Rabbit Tool (Forcible entry)	Radio Vehicle Stabilization Cable cutters/ wrench	Radio Rescuer Harness Patient Packaging Equipment
Back up					
Driver Radio ID= Rig# D/O	Radio 2 flashlights Turn table	Radio 2 flashlights SCBA Can Hook Ladder (roof/ egress)	Radio 2 flashlights Turn table/ ground ladders	Radio Power Unit Traffic safety Dry Chem Extinguisher	Radio Rescuer Harness Handwear Bag Rope Bag

Section VII: Building Construction

Overview

The firefighter's ability to ventilate a building safely and efficiently will depend on the firefighter's understanding of building construction. Construction methods and materials have changed significantly over the last several decades. It is essential that firefighters become familiar with the existing and newly constructed buildings within their response district. The following descriptions of both conventional and lightweight construction are not to be construed as complete or absolute. Basic characteristics, strengths and weaknesses are given to provide a basic framework of knowledge from which to operate safely.

Conventional Construction

Conventional construction gets its strength from actual size or mass. There is less surface area exposed to air or fire. There is more mass or fuel to consume, creating a longer burn time and a greater window of safety for the firefighter with respect to time. Roof framing components are continuous lengths of full-sized lumber. Ridge beams are single members with conventional rafters running from ridge to top plate. Rafter size will vary depending on span, pitch, and load. Spacing is usually 16" to 24". Additional members usually can be found in the form of collar ties and knee braces. Conventional sheathing material is most commonly 1' x 6' laid at 90 degrees to support members and spaced for shingles or laid at a 45-degree angle for support with no spacing. You will also find plywood used as sheathing in varying thicknesses. Conventionally constructed commercial buildings built during the 1930's and 1940's commonly used truss construction. Although the conventional truss's members have the same strength interrelationship, it is much stronger than its lightweight counterpart. This type of construction used 2' x 12' lumber for the top and bottom chord with rafters 2' x 10'. This type of construction is very strong, and early structural collapse is not an immediate concern.

Lightweight construction

In today's world, lightweight construction is predominantly used in the construction industry. With high labor and material costs, lightweight construction uses less lumber and smaller, low-cost members. In modern construction, laminated beams, heavy timbers and 1" x 6" sheathing have given way to 2" x 3" and 2" x 4" lumber and 1/2" plywood, regardless of building size. From a firefighting perspective, the use of less fire resistive materials translates to less time available to ventilate before the roof becomes unstable. This discussion will focus on the four major types of lightweight roof construction: panelized, open web truss, metal gusset plate truss, and wooden "I" beam.

Metal Gusset Plate Trusses

This type of roof system commonly found in residential and commercial buildings is usually 2" x 4" lumber butt jointed and held together by metal gusset plates, commonly known as a gang nail. The gang nail commonly penetrates 3/8”. Trusses are characterized by a top and bottom chord in tension and compression. The strength of the truss lies in the geometric interrelationship. Failure occurs when one component of the truss is consumed by fire, or the gang nail pulls loose due to charring. The most common spacing for trusses is 2' on center and the point where the truss crosses the bearing wall is the strongest location.

Wooden “I” Beam

This type of roof and sometimes floor system has a top and bottom chord of 2" x 3" or 2" x 4" lumber. The stem is normally 3/8” plywood or particleboard glued in place. Common spacing is 2’ on center, and the area where the roof meets the exterior wall is the strongest location. The stem has very little relative mass and burns to failure quickly.

Open Web Trusses

This type of system has a wooden top and bottom chord that are cross connected by steel tube web members. The top chord is in compression and the bottom in tension. The steel tubes have the ends pressed flat in a semicircular shape with a hole punched through them. The tubes are placed in slots in the chords with pins driven through them. The top chord usually rests on the bearing wall and the bottom chord is unsupported. Spans of up to 70’ are possible, normal spacing is 2' on center, and the area where the roof meets the exterior wall is the strongest point.

Panelized Roofs

The panelized roof normally consists of large, laminated beams spaced every 12” to 40" spanning the length or width of the building. They are supported by pilasters or steel posts on the ends. Along the span you will find either wooden or steel posts as supports. Beams can span well over 100' and are often bolted together. Normally purlins are installed with metal hangers on 8" centers perpendicular and between the beams.

Wooden 2" x 4" rafters are installed with metal hangers on 2’ centers, perpendicular to and between the purlins. Decking is usually 1⁄2" plywood. The safest and strongest locations are the beams, purlins, and perimeter of the building. The inherent weakness is the lightweight construction between the major framing members.

Roof Styles:

There are three basic categories of roof design: pitched roofs, arched roofs, and flat roofs. The following discussion identifies them by category and evaluates some of the more common styles with respect to strengths and weaknesses.

Pitched Roof Styles

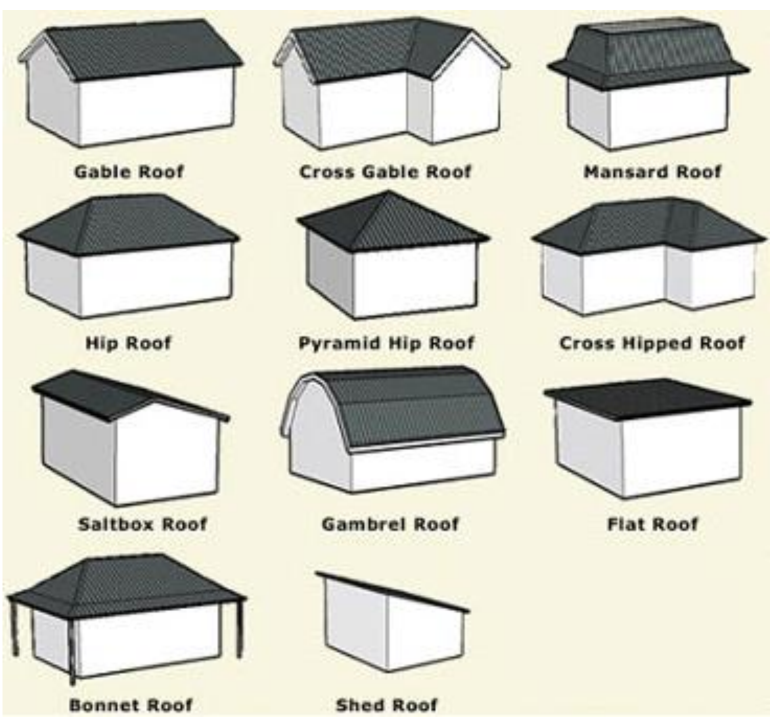
Gable: Basic A-frame design with the roof pitched in two opposing planes. If constructed in a conventional manner, the continuous ridge, exterior and bearing walls are normally the safest locations. In the lightweight version, the ceiling joists are the bottom chords of the truss, often not tied to the interior walls.

Hip: Two sets of opposing pitches where the roof slopes down to meet every outside wall. Strengths lie at the ridges, valley rafters and at the point where the rafters cross the outside walls. Weaknesses are the same as a gable when in the lightweight version.

Shed: Basically, this style is half a gable. The weakness here is the mono-pitched truss with a single web member subject to early collapse.

Bridge Truss: This is heavy duty trussing with sloping ends. The two parallel chords are inconstant tension and compression and can fail during heavy fire exposure. However, the likelihood of failure is dependent on the dimensions of the materials and the span of the trusses. This roof usually fails in sections and may have a large open attic space.

Mansard: The mansard roof has a double slope on each of its four sides. The lower slope is steeper than the upper slope. The four sides meet in the middle in a hipped peak/ridge. If it is a more modern version, the sides form a central flat area. This type of roof is usually bridge truss construction and creates large dead spaces and a potential for early collapse.



Arched Roof Styles:

Bowstring: The arched chords are usually 2"x 12” lumber with 2” x 10” rafters. Tie rods with turnbuckles are used for lateral support and to regulate tension. The roof is quite strong, but collapse can occur if the tie rods are heated to failure.

Ribbed Arch: Construction is like the bridge truss except that the top chord is arched. The heavy timber is very fire resistant but is often open with no attic to protect the framing.

Lamella Roof: A geometric egg crate or diamond pattern frame with sheathing laid over it. The 2” x 12” members are bolted together with gusset plates. The roof is supported by exterior buttresses or internal tie rods.

The Flat Roof:

Conventional flat roofs are constructed with rafters 2” x 6” or larger depending upon the span. Rafters are covered with 1" x 6" sheathing often laid at 45 degrees to the outside walls. The perimeter of the building where the rafters rest on the exterior wall is considered a strength. Due to mass, rafters are also considered safe locations. Several lightweight roof systems are used to produce flat roofs. The wooden "I" beam, metal gusset plate truss, and open web types are all previously mentioned possibilities. Identification of building construction type is most difficult in flat roof buildings. The single best way of being certain about building construction type is having previous knowledge of the building.

<u>On scene roof reports:</u>

With all this information you can size up a roof and give a roof report to on-scene crews and command. A roof report should be short, concise, and efficient describing unexpected hazards that are encountered.

The amount of information will be dictated by the extent of the incident.

Examples of things to look for, but not necessarily needed to be transmitted if no hazard present:

- Identify hazards if any. (Smoke, fire, stability, etc.)
- Is there an active fire in the attic?
- What is the roof load and roof type; for example, are there A/C units, evaporative coolers, skylights, or storage items, and is the roof flat or built-up?
- Roof integrity—is it strong or are there any sagging or spongy roof areas?
- Other factors, such as firewalls or hazards
- Announcement of the retreat off the roof, if applicable
- Report to command when back on the ground, followed with a personnel accountability report (PAR) if applicable.

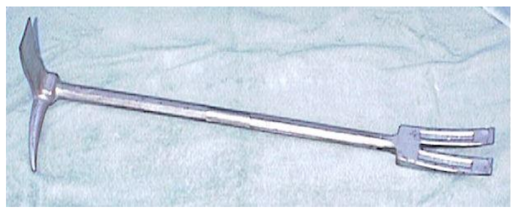
Following is an example of a good roof report: “Command from Truck 8 with a roof report. There is light lazy smoke from a flat composite roll top roof with a firewall to the Bravo side that appears to be intact. The roof has good integrity, with a normal load of six industrial-sized air conditioning units and skylights. We will be preparing for vertical ventilation.” Consider no good news reporting. If the incident dictates another acceptable roof report is “Roof is clear”.

Section VIII: Forcible Entry

"Try before you pry"

Overview:

Fire cannot be extinguished, searches cannot be made, and extension of fire cannot be checked until entry is made. The primary motivation should be professionalism. When forcing entry, you have an obligation to get the job done safely, efficiently, and with the least amount of damage possible. To accomplish this task, there are an assortment of tools and techniques that can be utilized. Consideration should be given to performing forcible entry with the least amount of damage. However, there are times when brute force must be combined with skill, technique, and knowledge to gain entry. The fire fighter assigned the job of forcible entry is given these responsibilities.



Forcible Entry Tools:

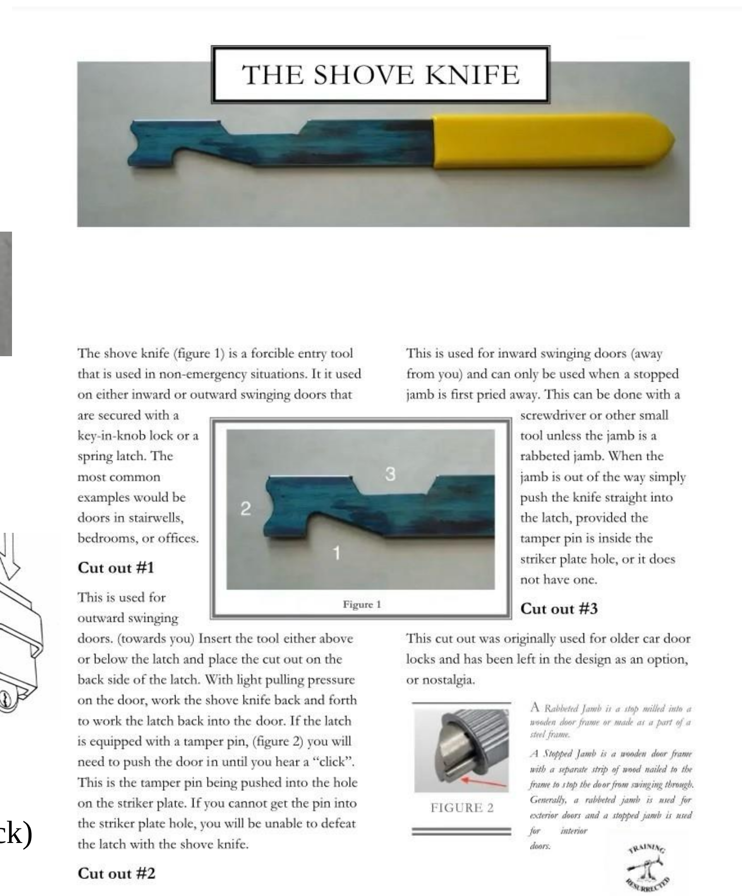
Conventional Tools

Axe (6 and 8 pound)

Halligan Tool

Pig (10 pound)

NY Hook (steel shaft)



THE SHOVE KNIFE

The shove knife (figure 1) is a forcible entry tool that is used in non-emergency situations. It it used on either inward or outward swinging doors that are secured with a key-in-knob lock or a spring latch. The most common examples would be doors in stairwells, bedrooms, or offices.

Figure 1

Cut out #1

This is used for outward swinging doors. (towards you) Insert the tool either above or below the latch and place the cut out on the back side of the latch. With light pulling pressure on the door, work the shove knife back and forth to work the latch back into the door. If the latch is equipped with a tamper pin, (figure 2) you will need to push the door in until you hear a "click". This is the tamper pin being pushed into the hole on the striker plate. If you cannot get the pin into the striker plate hole, you will be unable to defeat the latch with the shove knife.

Cut out #2

This is used for inward swinging doors (away from you) and can only be used when a stopped jamb is first pried away. This can be done with a screwdriver or other small tool unless the jamb is a rabbeted jamb. When the jamb is out of the way simply push the knife straight into the latch, provided the tamper pin is inside the striker plate hole, or it does not have one.

Cut out #3

This cut out was originally used for older car door locks and has been left in the design as an option, or nostalgia.

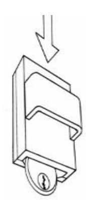
FIGURE 2

A Rabbeted Jamb is a stop milled into a wooden door frame or made as a part of a steel frame.

A Stopped Jamb is a wooden door frame with a separate strip of wood nailed to the frame to stop the door from swinging through. Generally, a rabbeted jamb is used for exterior doors and a stopped jamb is used for interior doors.

TRAINING RESURRECTED

Thru-the-Lock Tools



K-Tool

Lock Puller (Rex Tool)

Shove Kinfe

Vice Grips (may be used for Padlocks, Thru-the-Lock)

Hydraulic Tools

Hydra-Ram

External Lock Tools

Duckbill Lock Breaker

Bolt Cutter

Power Tools

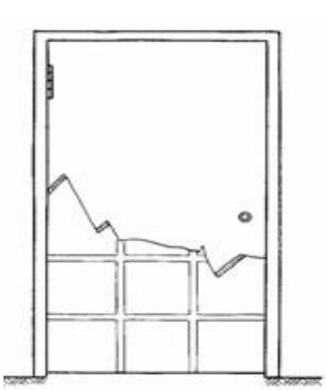
Rotary Saw

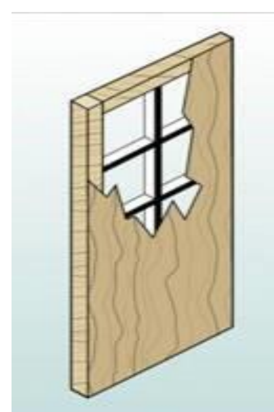
Cordless Drill/Cordless Sawzall

TYPES OF DOORS

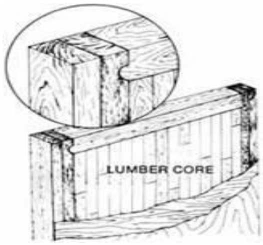
WOOD DOOR - There are two types of wood doors: Hollow Core and Solid Core.

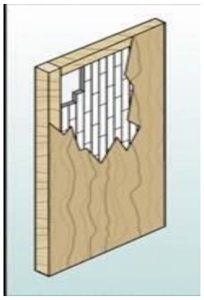
Hollow Core: Made up of an assembly of wood strips formed into a grid. These strips are glued together within the frame forming a stiff and strong core. Over this framework and grid are layers of plywood veneer paneling.





Solid Core: The entire core of the door is constructed of solid material such as tongue and groove boards that are glued within the frame. Other solid core doors may be filled with a compressed material that is fire retarded. In either case, the door is sided with a plywood veneer covering.





METAL DOOR

Constructed of metal, these doors are usually set in hollow or filled metal doorframes. When set on a masonry wall, as well as a metal frame, they are quite formidable and will hold back considerable fire. Today a metal door is quite common even in private dwellings.

TEMPERED GLASS DOOR

Distinguishable by the lack of a full frame with little or no trim. The door handle is usually mounted through glass. The lock may be installed in either the top or bottom style, **usually the bottom one.** Commonly known as a "Glass Door."

The breaking characteristics of Tempered Glass are quite different than ordinary Plate Glass. This is due to the heat treatment given to the glass during tempering. This results in high-tension stress in the center of the glass and high compression stress in the exterior surfaces. These tension and compression stresses balance each other. The heat treatment also increases strength and flexibility as well as the resistance to shock, pressure, and temperature increases. Approximately four times stronger than plate glass, when broken, tempered glass disintegrates into relatively small pieces.

ALUMINUM FRAME GLASS DOOR

These are the most popular doors in commercial occupancies, especially the taxpayer type. It is not uncommon to have the plate glass replaced with tempered glass, lexan or plexi-glass in some areas.

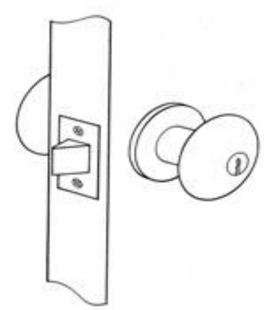
SLIDING DOORS

These doors may travel either to the right or left of their opening or in the same plane.

Sliding doors are usually supported upon a metal track and their side movement is made easier by small rollers or guide wheels. A bar may be placed between the fixed frame and the door or in the track to prevent unlawful entry.

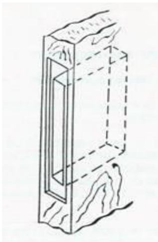
TYPES OF LOCKS

KEY-IN-THE-KNOB LOCK - As the name implies, the locking mechanism is part of the knob. These locks are found on both residential and commercial doors.



TUBULAR DEAD BOLT - This is a very popular locking device. It may be single or double key activated. It is a cross between a mortise lock, rim lock and a key-in-the-knob lock.

MORTISE LOCKS - Are designed and manufactured to fit into a cavity in the edge of either a metal or solid wood door. They have a solid, threaded key cylinder, which is secured in place by setscrews. The two most common types are Mortise/Latch Key and Mortise/Door Knob (see below).

ADDITIONAL SECURITY DEVICES

Static Bar/ Drop Bars

A fastening device that can be mounted across the door at any point. Generally, they are in pairs. The bars are held in place by brackets, which may be fastened to the doorframe.

NOTE: With the Sliding Bolt and Static Bar in place, you know the occupants did not exit through that door. There is either another means of egress, or the occupants are still inside. Static bars in place may not be visible from the outside.

CYLINDER GUARDS

A raised rectangular metal plate over the lock cylinder that is held in place with four carriage bolts. These bolts may be exposed or hidden in the body of the guard.

Black **Heavy-Duty** **Brass**

CONVENTIONAL FORCIBLE ENTRY

Prior to forcing a door: The Forcible Entry Team should: **TRY THE DOOR** to determine "**IS THE DOOR LOCKED?"** Too many times over-aggressive firefighters have forced an unlocked door. They should take note of the **Type of Door and the Locking Devices** involved. Also, what are the **Prevailing Conditions** at the scene, such as heat, smoke, and visibility? They should then feel the door and or the doorknob. This may give an indication of the amount of heat behind the door. Finally, **Check for Resistance**; push in at top, center, and bottom of door. This may give you an idea as to where the locking devices are secured.

STEPS FOR FORCING A DOOR

The recommended steps for forcing a door are **GAP – SET – FORCE.**

FORCING *INWARD* OPENING DOORS (door swings away from you)
GAP the DOOR - This step will make an opening in the door and/or frame to create a purchase point. It may also force open a poorly secured door.

Work the **ADZ** into the stop on the doorframe approximately 6 inches above or below the lock (see "Note" below). The tool can be set into the frame by swinging like a bat and driving the **ADZ** into the frame.

If there are 2 locks close together, go between them (unless they are stacked locks).

Push up or down on the Halligan Tool causing the **ADZ** to rotate and crease the door. The best purchase is gained when the **ADZ** end is used on the door**,** not the pike.

Note: The reason for the 6-inch rule is to prevent the Halligan Tool from striking the lock. The fork of the Halligan Tool is approximately 3-inches wide, and most lock bodies are also 3-inches wide.

Technique Tip: You will lose power when pushing down if the pike hits the door. You will increase spread by moving the tool up.

SET THE TOOL - This step requires the most skill. This involves working the **FORK** of the Halligan Tool into the **Gap** to spread the door away from the frame. The Halligan Tool is considered “**Set**” when the **FORK** is “**locked in**” to the inside of the doorframe.

Position the Halligan Tool **FORK approximately six inches above** or below the lock cylinder. If the tool is too close, the **FORK** may hit the lock and will not go through to “**lock in**.” If it is too far away, the door may flex, and the lock will not fail.

Place the **FORK** of the Halligan Tool and angle the Halligan Tool to work around the doorstop. This is considered the ideal position since it produces the most spread of the door and frame and puts the most stress on the locking device. It is important for the member holding the Halligan Tool to “**walk the tool**” around the doorstop and frame.

This method gives a greater range of motion to the Halligan Tool since the **adz** will be facing **away** from the door and not strike the door when the door is forced.

It also offers a better striking position. The Halligan Tool will stand out at approximately 90 degrees to the door allowing the member with the axe more room to maneuver and deliver the necessary blows.

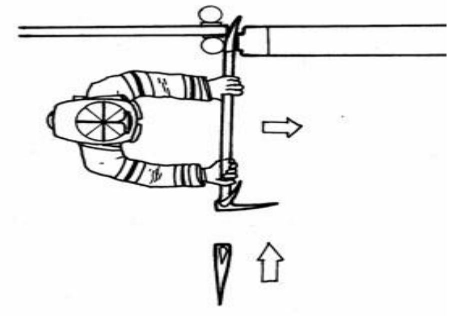
Note: When there are multiple locks closely spaced on the door (stacked locks), position the tool above the upper lock or below the lower lock. Remember the six-inch rule is a general rule and should allow the FORK to clear the inside to the lock.

The **forcible entry firefighter should be between the door and the tool.** Generally, the forcible entry member should have his shoulder in contact with the door. This position gives a good view of the area where the tool is being driven in and gives full range of motion for the tool as it is pushed away from the door as it is being driven in.

The forcible entry firefighter should keep his **eyes on the FORK end** of the Halligan Tool where it is being driven into the **Gap.**

Keep moving the Halligan Tool away from the door as it is being driven in (struck).

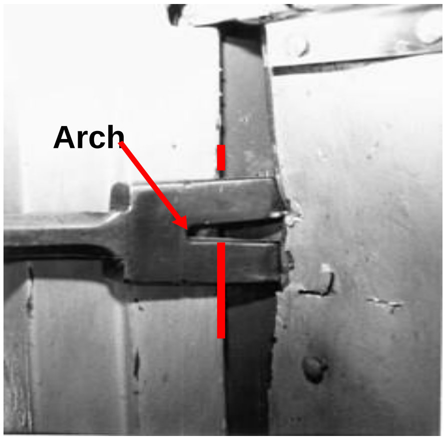
SET THE TOOL

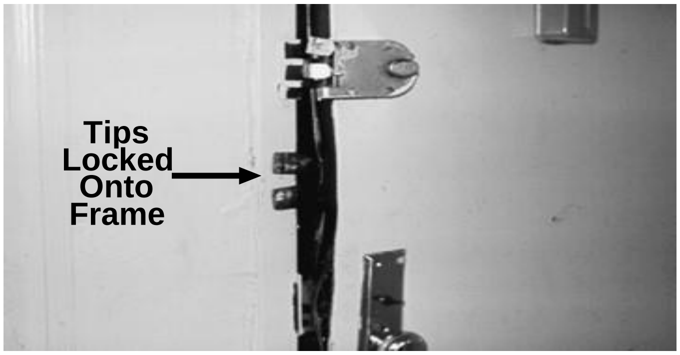


Technique Tip: As soon as the tip of the fork is past the edge of the door, sharply push the tool away from the door. "Spring" the door away from the frame and maintain pressure on the tool to prevent the tips from striking the frame.

When the Halligan is nearly **perpendicular** to the door, **drive in forcefully.** The **FORK** end of the tool is driven past the inside of the frame. This will ensure the tool being "**locked**" into position and not slipping when pressure is applied.

The tool is **SET** when the **ARCH of the FORK is even with the inside edge of the door / doorstop.**





STRIKING THE HALLIGAN TOOL - Coordination and communication must be maintained between the members of the forcible entry team.

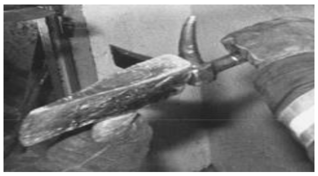
The member holding the Halligan Tool (forcible entry firefighter) controls the operation.

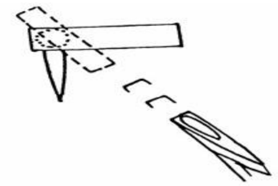
The member with the axe strikes the Halligan Tool **PERPENDICULAR TO THE ADZ.**

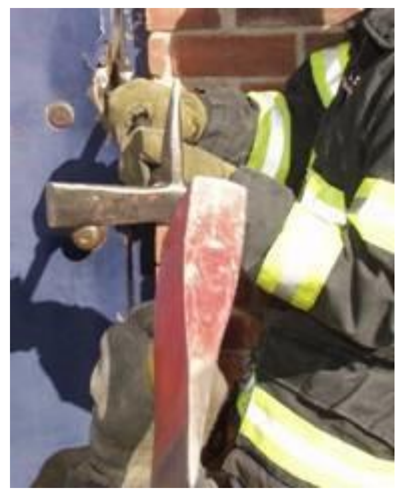
The member with the axe may have to stand, crouch or kneel to obtain the best position.

The member with the axe strikes the Halligan only when told.

The commands "**HIT**" and "**STOP**" must be understood.







TO MAINTAIN CONTROL

- Short chopping blows.
- Perpendicular to the adz.
- In line with the shaft.

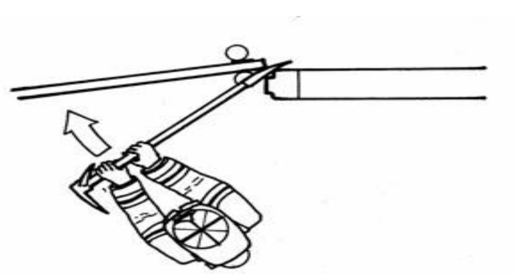
Note: As the tool is SET, more powerful blows can be delivered.

FORCE - When the Halligan Tool is set, **force** is applied to the tool creating leverage against the door.

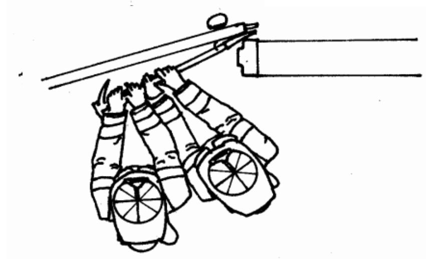
Forcible entry member **changes position to Face the Door.** This gives him a better position to apply pressure.

Ensures everyone is **ready.**

The other member of the team should try to control the sudden opening of the door by holding onto the doorknob or applying a hose strap to the knob.



Push in sharply to create maximum force.



If strong **resistance is met**, a second firefighter may be used to **assist.**

As the door opens, the second firefighter must **MAINTAIN CONTROL OF THE DOOR.**

Note: In the above method, as the door is flexed from the pressure, note the presence of fire behind the door. If fire is present, make sure there is a charged line in position to protect the forcible entry team.

ALTERNATE METHODS TO GAP AN *INWARD* OPENING DOOR

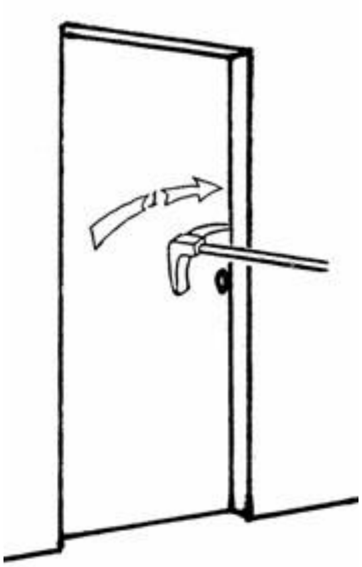
Pike or Adz into the Frame

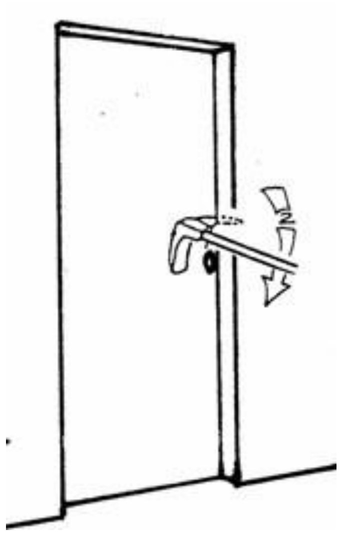
Driving the **PIKE or ADZ into the doorframe** with either the axe or maul, or simply by taking a "baseball-bat swing," should give the tool enough bite to ensure a purchase. Try to bury the **PIKE** into the frame as close to the door and lock as possible. This procedure is very quick and simple for a one-man operation. **This procedure may force the door.** It works best on wooden doors with wooden frames.

Place the **PIKE** between the door and the doorstop, on or near the lock.

Drive (set) the **PIKE** with the axe.

Push down or Pull into the door (Gap) with the Halligan Tool.





Bevel to the Frame

- Place the **BEVEL** of the Halligan Tool **against the frame** and with an axe or maul, drive the Halligan Tool in.
- This is usually done when there is a very tight door with stiff resistance:
- Usually, a metal door with a metal frame, or obstruction is in the way making it difficult to strike the tool.
- As the tool is driven in, it must not be driven **into the frame**. This takes a "**fee**l" of the tool to do correctly.

Note: This method does not give the full range of motion to the tool. The ADZ will strike the face of the door as the member pushes towards the door.

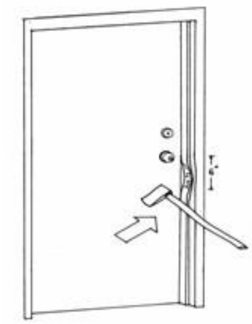
Shocking the Door

Shocking the Door with a few sharp blows with the Halligan Tool, axe or maul to loosen the door to allow the adz to slip in. However, when using this method, you must hit the **"rail of the door,"** since this is usually the strongest area of the door.

Technique Tip:

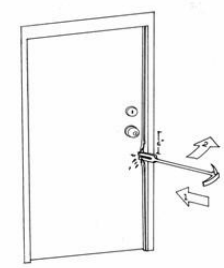
If the door is set in a weak wood frame, several **sharp blows to the door right on the lock** may split the frame. This is especially true if the door contains a **mortise lock**. Note the mortise lock is set into a cavity made in the door. This may compromise the integrity of the door.

Batter the Door Frame



Batter the Door Frame by striking with an axe, maul, or Halligan Tool approximately 6 inches above or below the lock and driving it away from the door to allow entry for the Halligan Tool. Sometimes steel frames are filled with concrete and may not be crushed.

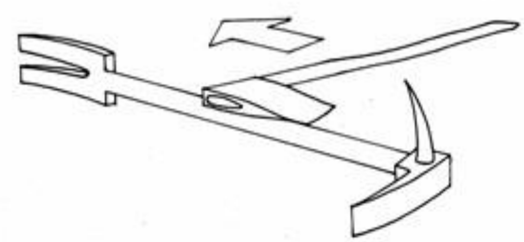
Remove the Door Stop



Remove the doorstep on wood doors with the **ADZ** or **FORK** end of the Halligan Tool. This is a simple way to open a door with minimal damage. This method works best on wood doors with wood frames.

NARROW HALLWAY OR RECESSESED DOOR

When there is little room and/or limited visibility to swing an axe, an alternate method of striking the Halligan would be:



Slide the head of the Axe **along the shaft** of the Halligan Tool and **strike the shoulder** of the fork end.

Bevel to the Frame may give better results due to easier entry and better angle to strike the tool.

Note: The shoulder of the fork end of the Halligan must be squared off to do this.

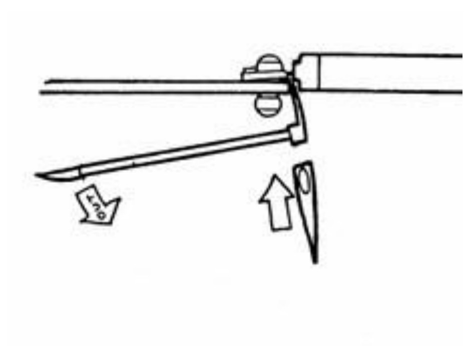
<u>FORCING OUTWARD OPENING DOORS (Door swings toward you)</u>
Using The Adz End

Place the **ADZ** between the door and the frame.

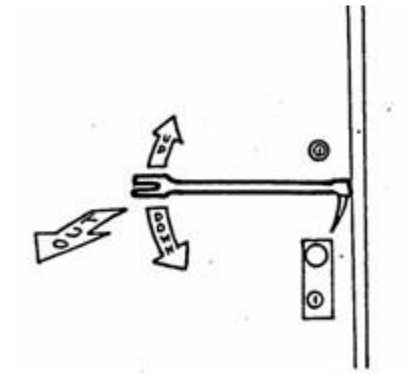
GAP the door by rocking the tool **up and down** to spread the door from the frame.

SET the tool, and pry the door out by **pulling** on the Halligan so the **ADZ** can be driven in. Be careful not to **"bury the tool"** into the doorstop. See Chapter 16.

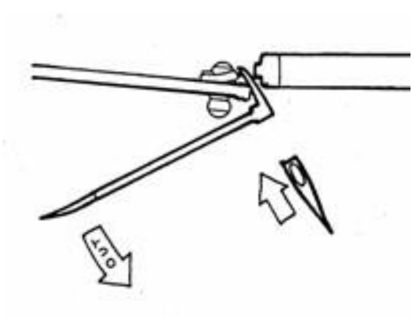
Force the door, set the **ADZ** end around the inside of the door.



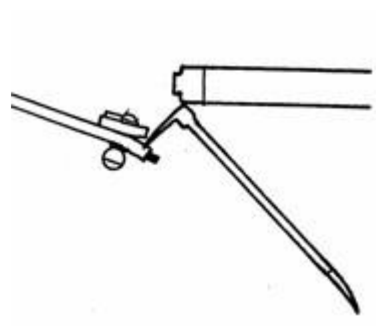
Gapping The Door (Top View)



Gapping The Door (Front View)



Set The Tool (Top View)



Force The Door (Top View)

Note: The firefighter always faces the door.

FORCING OUTWARD OPENING DOORS (Door swings toward you)

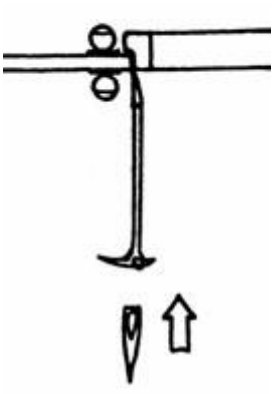
Using the Fork End

GAP the door by placing the bevel side of the **FORK** toward the frame, just above or below the lock or hinge.

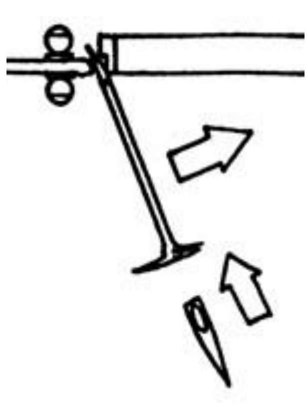
SET the tool, pry the door by **pulling** out on the Halligan so the **FORK** can be driven in past the inside frame. Be careful not to **"bury the tool"** into the doorstop. See Chapter 16.

FORCE the door, set the **FORK** end around the inside of the door and by **pulling or pushing** the Halligan Tool ***away*** from the door (toward the wall).

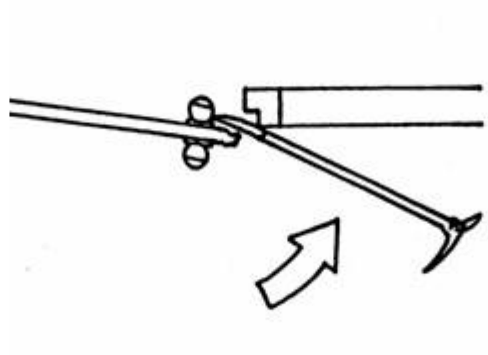
To use this method, the Halligan Tool must have sufficient room to allow the movement of the tool away from the door.



Gap (Top View)



Set (Top View)

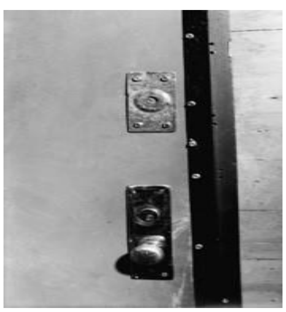


Force (Top View)

Note: These methods will be dictated by the configuration of the building or any obstructions near the door.

METAL STRIP ON THE EDGE OF THE *OUTWARD* OPENING DOOR

Additional security may be installed on these doors by bolting a metal shield to protect the space between the door and the frame. It may be a full-length or partial shield. Dealing with the shield will require an additional step before proceeding to **Gap – Set – Force.**

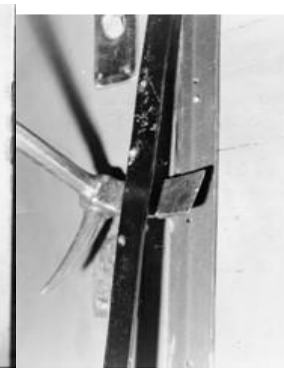


- Drive the **ADZ** end under the edge of the metal strip and push the tool toward the door.
- Work the **ADZ** between the door and the frame and drive in to establish a gap.

- Drive the **FORK** end under the edge of the metal strip and push the tool back toward the door.
- Work the **FORK** between the door and frame. Reverse the tool if necessary.



- Drive the **ADZ** end between the door and shield, bending the **shield** away to allow entry of the Halligan Tool.
- Shear the bolts and pry, bend or remove the shield as a **last** resort.

Note: At times if the angle iron is secured well, it may assist you in opening the door. If not, then you have to remove it to access the door.

HYDRAULIC TOOLS

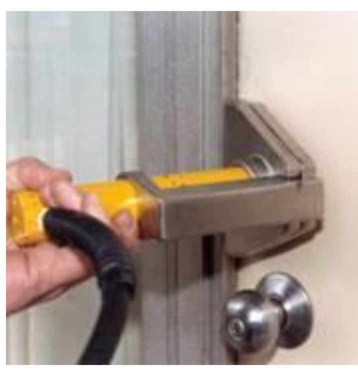
These tools are designed for **doors that open inward** (away from you), and have also been used successfully on **sliding elevator doors**. They work best on doors with **strong metal frames**.

RECOMMENDED STEPS FOR FORCING A DOOR

GAP THE DOOR – Using the **ADZ** end.

SET THE TOOL - Insert the jaws between the door and the frame midway between the knob and the lock; the **jaws must be in the closed position.**

FORCE THE DOOR - The door should open with several pumps of the handle.


Gap Set Force

Note: When there are multiple locks, insert the jaws between the knob and lock and then move to the proximity of the next lock.

CHOCKING THE DOOR

This is a very basic and important task that gets overlooked from time to time. Many doors are self-closing and if not chocked open it delays other members from entering the fire building (occupancy). Whatever means used to chock the door must be **"positive,"** not something that can be knocked out unintentionally. However, it must be something that **can be removed quickly** if necessary. It is suggested that the first unit to enter the fire building be responsible for **"chocking"** the door. That could be the Officer or any member of the forcible entry team.

Some suggested methods of chocking a door:

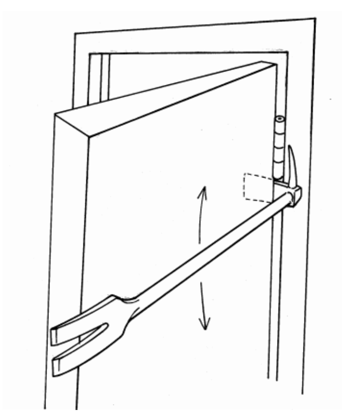
A wooden chock wedged under the door. Every member should carry at least two wood chocks in their pockets.

The head of axe slid under the door. As the forcible entry team enters the occupancy, the axe is wedged under the door. This method marks the door, keeps it open and safeguards the axe since it is rarely used **INSIDE** once the door is forced. If you feel the axe might be needed **INSIDE,** then this method would not be appropriate.

Placing the hydraulic tool against the open door. Once the door is forced, the hydraulic tool is not needed and can be used to keep the door open.

A hook chock placed over the hinge.

Disable the door by placing a tool between the frame and the door just above or below the hinge and prying down will break most hinges and keep the door open.



THRU-THE-LOCK ENTRY

The "Thru-the-Lock" approach is a means of gaining entry by attacking the locking device and **opening the door with little or no damage to the door or frame.** This is a professional method of entry and serves as a good public relations tool. In most cases, this method would only be used when **time and fire conditions are not urgent,** or where conventional methods would cause more damage than the fire itself. Examples would be high-rise office buildings, hotels, motels and/or commercial occupancies, where many rooms and or occupancies must be checked without causing too much damage. The Thru-the-Lock method usually does not create as much of a security problem as conventional forcible entry method. There are times with certain types of locks that the Thru-the-Lock method of forcible entry may be a quicker, more efficient means of entry, whatever the conditions. Since security has been improved through technology over the years, this book can only address what is most common. This chapter of the book will outline some basic principles, methods and techniques used in **Thru-the-Lock Entry.**

Size-Up

Though it is impossible to know for sure what type of lock is securing the occupancy by looking at a solid door from the outside, we can make an educated guess based on:

- Type of **occupancy.**
- Type of **door.**
- **Location** of the **lock cylinder(s).**
- **Direction** the door **moves** (inward or outward).
- What we **see on the door** (other than the locks).
- Anything **unusual** (lock cylinders out of line).
- Knowledge of the **type of lock.**
- Let the **fire condition** dictate your method of entry.

Combine all this information with experience and proceed in **attacking the lock,** not the door. We need to understand that only **practice** will make us more proficient in our operation, so we must use every opportunity.

Note: The cheaper the lock, the more difficult it may be to force. Cheaper locks tend to break up causing delays, and/or requiring alternative means of pulling the cylinder.

KEY-IN-THE-KNOB LOCK

As the name implies, the locking mechanism is part of the knob. These devices are found on both inward and outward swinging doors. The spring latch on the majority of these locks enters the striker approximately 1/2 inch.

FORCING THE KEY-IN-THE-KNOB LOCK - Using the Officer's Tool

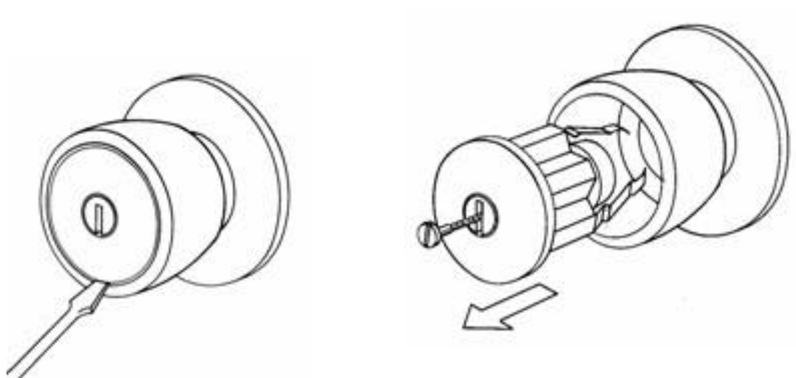
The doorknob can be removed simply and quickly with the Officer's Tool, without damaging the stem assembly.

If the **door is hollow,** an axe can be placed behind the tool to give the fulcrum a substantial base to pivot off.

After the doorknob is removed, insert the stem of the Key Tool into the slot (if present) or into the back of the spring latch and pull or twist toward the hinge side of the door to activate the latch.

FORCING THE KEY-IN-THE-KNOB LOCK – Removing the Center of the Knob

There are some locks where the center of the knob can be removed (example, Kwikset type lock) with a knife-like tool or slotted screwdriver.



FORCING KEY-IN-THE-KNOB LOCKS – Outward Swinging Doors

Key-in-the-Knob locks on outward swinging doors have a simple spring latch which can be slipped back (opened) with a flat tool such as a shove knife**.**

At times there is a simple device known as an **anti-loitering pin**, which may be added to the latch. This pin prevents the insertion of the shove tool without moving this pin first.

TUBULAR DEAD BOLT

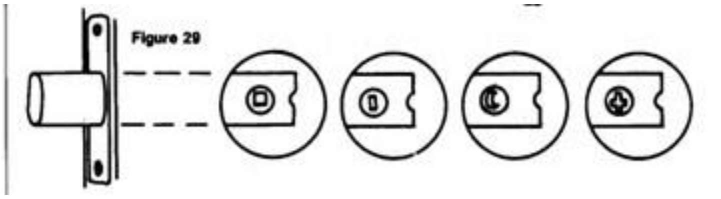
This is a very popular locking device. It may be single or double key activated. It is a cross between a mortise lock, rim lock and a key-in-the-knob lock. These locks may be recognized by their **position on the door and/or the size and shape of the cylinder.**

These locks have become more sophisticated as the demand for greater security has increased. They may have a hardened steel rod through the center of the locking bolt. The length of the bolt has been increased to the point that it **may take two full rotations of the key to remove the bolt from the keeper**.

The lock face is usually held in place by a hardened steel mounting. The cylinder is either **too deep or too wide**, which prevents the K-Tool from being used. To use the Thru-the-Lock method, the cylinder needs to be removed to enable the use of a Key Tool to trip the lock. If the K-Tool is unable to remove the cylinder, then an alternative method of removal would be needed to use this method.

If the cylinder is unable to be removed, then you will have to resort to conventional forcible entry methods to force the lock.

The stem of the tubular deadbolt, which retracts the locking bolt, can be in various shapes.



FORCING THE TUBULAR DEAD BOLT

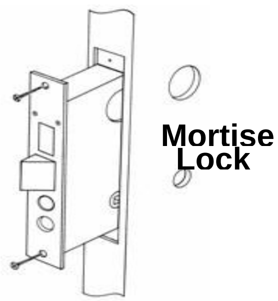
- Remove the cylinder by pulling it out with either the Rex Tool, K-Tool, or modified Halligan.
- Insert Key Tool.
- Rotate to open.

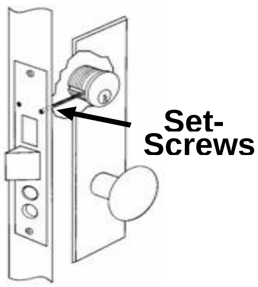
Technique Tip: Place the Rex Tool at an angle to start the operation.

MORTISE LOCKS

Are designed and manufactured to fit into a cavity in the edge of the door (either metal or solid wood). They have a solid, threaded key cylinder which is held in place by two setscrews. There are various types and styles of these locks available today.

A deadbolt and latch are a mortise type lock that contains both a latch and a bolt in one unit.





PRINCIPLE OF OPERATION – MORTISE LOCKS

As the key is rotated in the cylinder, it turns a cam on the back of the cylinder. This cam contacts a lever inside the lock box removing it from the strike. Although the key will cause the cam to make a complete revolution, the actual work of opening the bolt is usually accomplished between 5 and 7 o'clock or 7 and 5 o'clock of that revolution depending on which side (right or left) of the door the lock is mounted.

FORCING THE MORTISE LOCK:

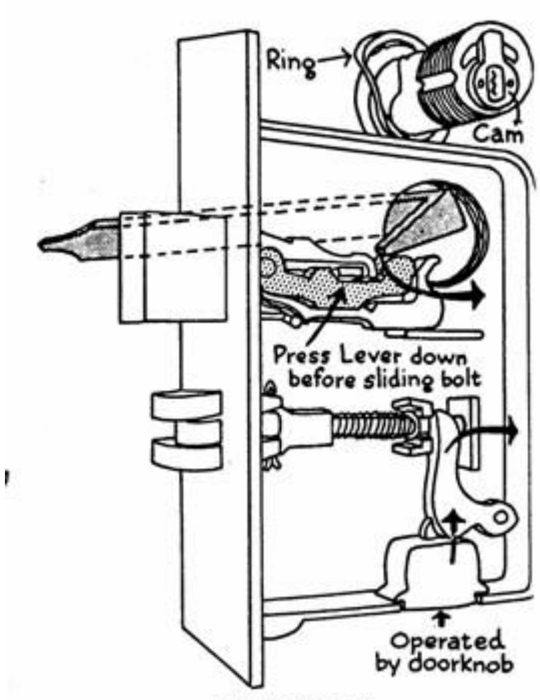
Set the K-Tool firmly on the cylinder and remove the cylinder by **pulling up.**

Insert the correct Key Tool.

Rotate the Key Tool. If the mechanism is found at 5 o'clock, rotate toward 7 o'clock, if found at 7 o'clock, rotate toward 5 o'clock.

If mounted with a doorknob, it may have a latch **that** may be connected to a second assembly. This may **necessitate a second revolution** of the cam **to** remove the cam from the keeper.

Note: Once you have pulled the lock cylinder be sure to use the proper end of the Key Tool.

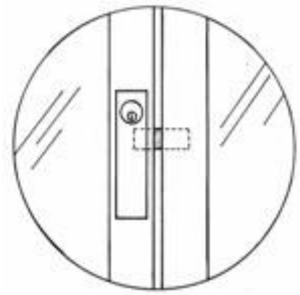


Lock Cylinder Removed Proper Key Tool

PIVOTING DEADBOLT

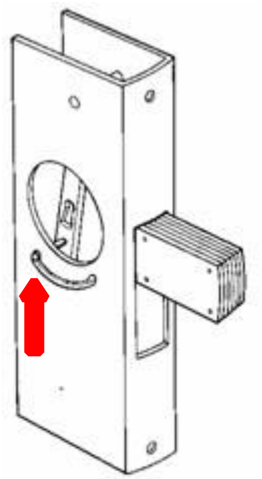
This popular lock is usually found on aluminum and glass panel doors with narrow stiles. It is also found on solid glass (tempered glass) doors with the frame on the top and bottom edges only. Generally, these are commercial occupancies.

PRINCIPLE OF OPERATION – PIVOTING DEADBOLT

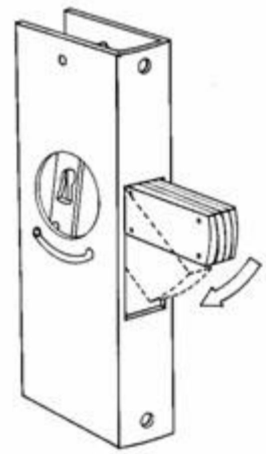


These locks usually have a laminated bolt, which may extend up to 1-3/4 inches. The tripping mechanism is slightly different than other mortise locks, which requires the correct Key Tool to be used to depress the locking pin, which rotates the dead bolt. The pivoting bolt allows forward throw to be the entire depth of the frame channel.

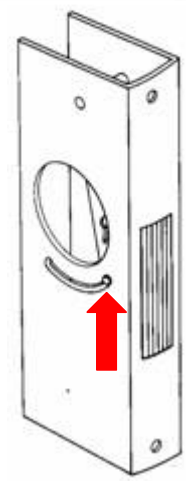
Pivots
Door
Door



Pin Away
Pin
Locked
Unlocked



Bolt
Forward
into Frame



The above is a narrow stile, pivoting deadbolt showing the 1¾ inch laminated bolt. The locking pin is **AWAY** from the leading edge of the door. The door is **locked** when the pin is in this direction. As it is depressed the bolt **"pivots"** into the frame. When the locking pin is **FORWARD**, the bolt is inside the frame and the door is **unlocked.**

FORCING THE PIVOTING DEADBOLT - Using The K-Tool

Pulling this cylinder is usually no problem for the **"K -Tool"** (it was designed for this lock). These cylinders rarely break apart.

FORCING THE PIVOTING DEADBOLT - Using The K-Tool

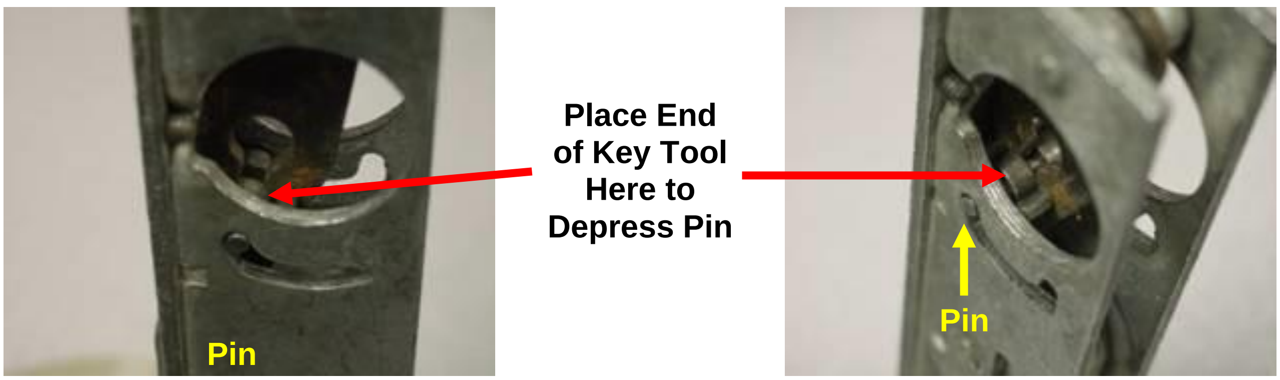
Place the K-Tool over the cylinder and set by driving down over the face of the cylinder to lock onto the cylinder.

Pry **UP** with the **ADZ** end of the Halligan, removing the cylinder.

Using the bent end of the Key Tool, **DEPRESS** the pin and **SLIDE** the pin forward, pivoting the deadbolt down into the housing.

As the locking pin slides forward, the bolt is retracted, unlocking the door.

Note: The pin will be located at either the 5 o'clock or the 7 o'clock position. Move the pin from 5 o'clock to 7 o'clock, or from 7 o'clock to 5 o'clock to retract the bolt, unlocking the door.



FORCING THE PIVOTING DEADBOLT – Using the Vise-grips

The cylinder may be able to be turned out of the lock by using a pair of vise-grips. Since all cylinders are held in place with setscrews, a quarter turn clockwise may bend the set screw just enough to allow you to **turn the cylinder counterclockwise** and remove it. After entry is accomplished, the cylinder may be screwed back into the lock box. This method may work on most mortise locks.

Note: If the cylinder guard is beveled or rotates freely, pulling the cylinder is a difficult, if not impossible, task.

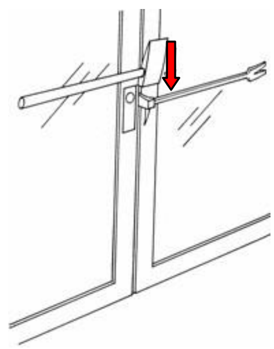
Using The Saw

If the occupancy has **center opening double doors,** take the **forcible entry saw** with the metal cutting blade and cut the bolt between the doors. There is enough room between the doors because of the door swing and the space is usually covered with only weather stripping. This may work with a single door if there is clearance for the saw to get in.

Using the Halligan

Place the **ADZ** end of the Halligan between the door and the jamb, with the bar of the Halligan in line with the cylinder, and parallel to the ground. Strike the Halligan with an axe or maul **DOWNWARD** on the **ADZ.** This may snap the pin holding the bolt and pivot the bolt out of the keeper.

This may work with single or double doors if there is room to place the Halligan.



PADLOCKS

Padlocks are detachable locking devices having a sliding and pivoting shackle that pass through fixed or removable hardware and then made secure.

Padlocks are used on both the exterior and interior of occupancies. They are found in the places you least expect, and you may have to force one with only the tools you carry. Therefore, members should be able to identify the various types of padlocks and their attachment hardware and means of installation.

CATEGORIES OF PADLOCKS

For size-up and understanding of padlocks, they are placed in three (3) categories:

Light duty.

Heavy duty.

Special purpose.

PADLOCK SIZE UP:

Type of padlock.

Hardware and installation (attachment device).

How many padlocks and their location.

Accessibility.

PARTS OF A PADLOCK

Shackle or bow.

Body, solid or laminated.

Keyway.

LIGHT DUTY PADLOCK

Shackle or bow is usually **1/4 inch or less.**

Shackle or bow **usually not case-hardened.**

Body of lock, solid or laminated.

Keyway (type may vary).

HEAVY DUTY PADLOCK

Shackle or Bow, **1/4 inch and larger.**

Body of lock, solid or laminated.

Case-hardened steel.

Toe and heel locking.

Guarded keyway.

FORCING PADLOCKS – Using the Forcible Entry Saw

When possible, use the cut off saw blade. This should be the **primary tool** to remove padlocks, hardware, and attachment devices. It offers speed and is relatively safer than striking tools.

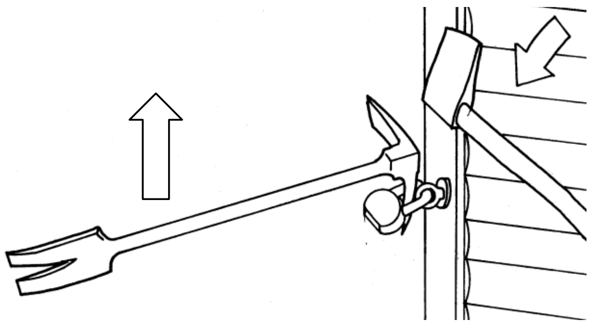
PIKE OF THE HALLIGAN TOOL

The **PIKE** of the Halligan Tool may be more effective on padlocks with short shackles.

Place the pike into the shackle opening, keeping the Halligan Tool as horizontal as possible.

Maintain pressure on the lock body.

Deliver sharp blows with a maul or axe.



BOLT CUTTERS

Bolt cutters are excellent for cutting light duty pad locks, light duty chains, cable and hardware. As a last resort they can also be used to cut heavy-duty padlocks, but when used this way, they may damage the jaws of the bolt cutter.

If they must be used for a heavy-duty padlock:

Open the bolt cutter to the maximum.

Position the bolt cutter so one handle is securely against a substantial object (wall, ground, etc.).

Push with both hands on the handle to cut the hardware.

Note: Most heavy-duty padlocks have toe and heel locking. Both sides of the shackle may have to be cut or twisted to remove the lock.

CUTTING THE ROLL-DOWN DOORS

There are many ways to cut roll-down security gates to gain access. There are just as many theories to justify these cuts. Each has its own merits but for simplicity, we are showing just a few.

Remember, each fire situation will dictate the appropriate cut.

Drop Cut

This operation requires 3 cuts and will remove a large portion of the door in 1 large piece. The key to this evolution is to ensure that both vertical cuts are through the bottom rail into the ground. Saw operator may need to adjust the blade guard on the rotary saw to achieve this. These vertical cuts should be approximately 1’ away from the left and right sides of the door. When performing the horizontal cut the saw can be placed on the shoulder to facilitate the cut to prevent arm fatigue. Be aware of the weight of the door that will be “dropping” to the ground and ensure the saw operator is out of the path of travel. Once the door is on the ground it can be left there or drug away to mitigate trip hazards.

Advantages:

3 cuts will remove most of the door.

A very large opening and egress path.

Can be performed with 1 member.

Can be utilized to remove slats on roll up curtain doors.

Disadvantages:

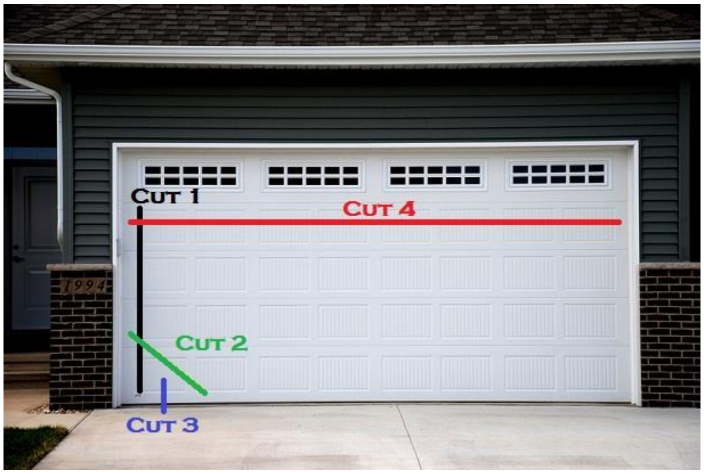
Large opening with uncontrolled flow path.

Horizontal cut can limit egress if not high enough.

Weight of door may introduce opportunity for injury.

Door *within* a Door Cut

This operation requires four cuts. The key to this evolution is ensuring the bottom rail has been cut clean trough. Unlike the drop cut where gravity brings the cut to the ground, here most of the doors are pulled to the side giving a wider access. (Note: If you can cut through the bottom rail with the first cut, cut 3 may not be needed.)



Advantages:

Less of a pile in front of the opening.

Can be used on very wide openings.

Can be partially closed back if needed to control flow paths.

If done correctly, the opening will be squared.

Disadvantages:

Requires more time.

Requires more than one member.

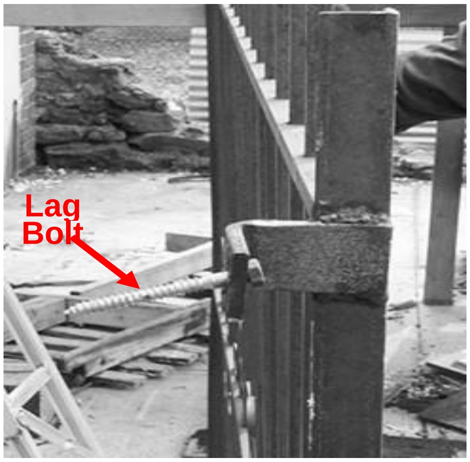
Window Bars

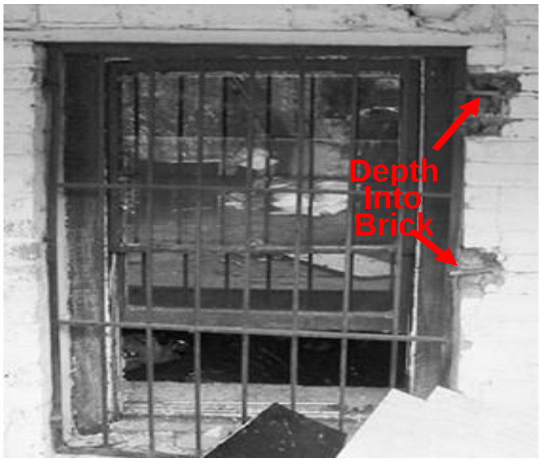
These obstacles come in a variety of sizes, shapes, and strengths. They may be mounted to the window frame with screws or bolts or set into the mortar. Bars and gates are used primarily for security and leave very little room for error in the case of fire.

Attacking and removing these obstacles during a fire situation takes time. If fire is being vented through the window being worked on, it becomes more of a challenge. Anyone trapped behind them has little chance of survival.

Forcing Window Bars

Bars are usually secured to a window at four points. The mounting point may be a lag bolt into the mortar or brick, or the mounting point may be part of the brickwork.

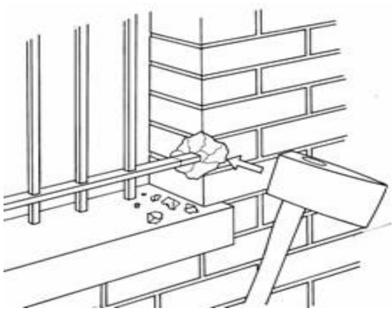
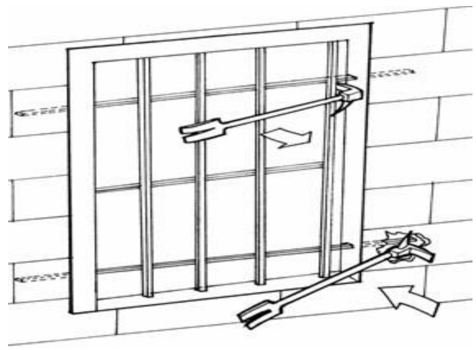




Forcing Window Bars

Bars are attacked at the weakest point. Striking the mortar or brick work where the bar is mounted may dislodge the anchor point. Dislodge two and bend the bars away or pry out using the Halligan.

Using a Halligan Tool, you may be able to pry the bar from its mount.

Rotary Saw

When using a rotary saw, cut the mounting bracket and remove the entire bar assembly, or cut two sides and bend the bars away.

Hydra- Ram

Place the Hydra-Ram between the angle iron outer edge of the window bar and the exterior of the building, as close as possible to the lag bolt, and spread.

If the spread is not sufficient, the tool can be repositioned directly at the point of attachment.


After forcing the attachment points on one side, push the bar assembly to the side (while still attached with a hook), allowing an unobstructed opening.

Pushing the gate to the side still attached will cause the gate to break free, dropping it to the ground.

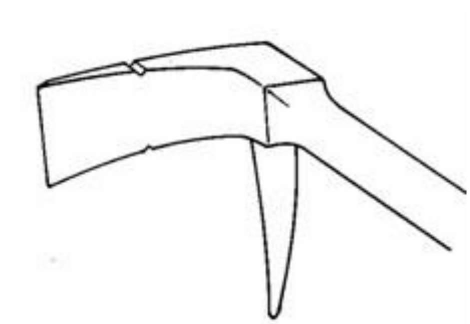
Keep this area clear to prevent anyone from getting hit by the falling gate.

Note: Start this operation from the bottom and work up to stay out of the path of the gate if it should fall.

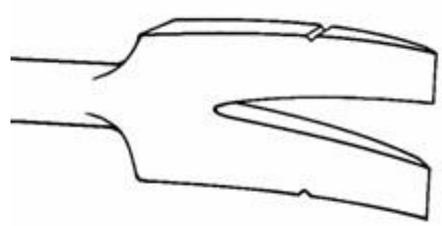
Halligan Tool Tips and Tricks

Marking the Halligan Tool for judging depth of the door when setting the tool.

Mark the adz and the fork with a notch approximately 1¾ inches up from the end to denote the depth of the door. When trying to “lock” the tool in, this will give you assurance that you are deep enough.



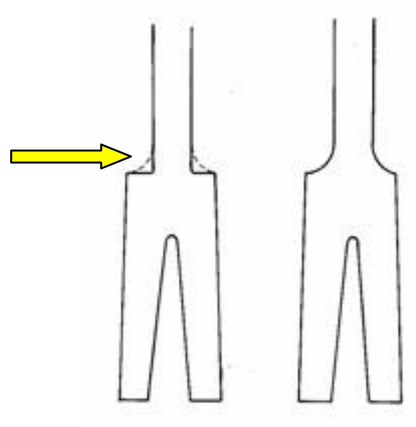
Adz End.



Fork End

Squaring the Shoulder

By squaring the shoulder of the fork end, it allows another striking surface in tight spaces and when the tool is recessed.

TIPS

Notching the Adz

This may give you the option of pulling a cylinder, getting a lock on, or shearing a bolt or screw. The edge is filed to a clean edge.

Section IX: Ground Ladders

Types of ladders

Straight ladder- single section ladder. Can include roof hooks and butt spurs on both ends. This ladder is not exclusive to roof work and can be one of the fastest used ladders if it is sufficient in height. (Current working lengths 14' and 16")

Extension ladder- multi section ladder consisting of a base, section fly section(s), pulleys, halyard, dogs/pawls. This ladder selection is adaptable and can be used appropriately at multiple heights. Ladders should be thrown with the fly out as per manufacture recommendation.
(Current working lengths 24, 28 and 35 feet)

Attic/Scuttle hole- Collapsible ladder designed for easy access in tight quarters. Excellent tool to carry on rescue calls, i.e., elevator and below grade. Excellent tool for scaling a fence when making access to alternate locations on the fireground.

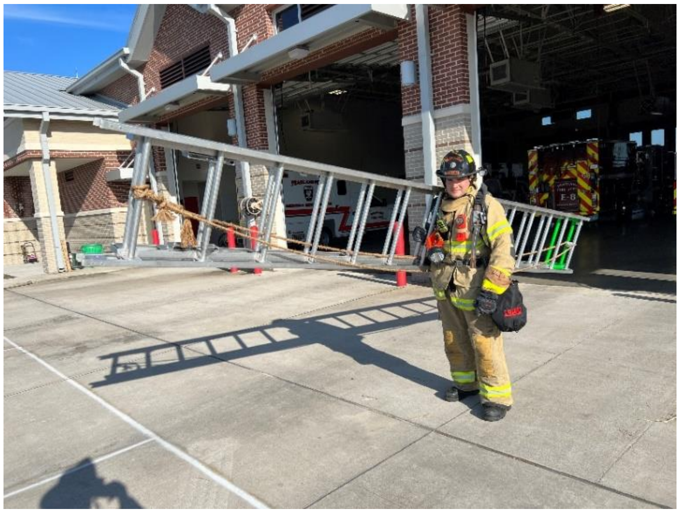
Little Giant- folding multi-use ladder that can operate as a straight ladder or a frame ladder. Excellent tool for good intent calls, overhaul, and extenuating circumstances.
The most forgotten tool on the truck.

ALL PORTABLE LADDER OPERATIONS ALWAYS REQUIRE A HELMET AND GLOVES AT A MINIMUM.

Carries

Ladders should be carried butt spurs forward to be thrown from the carry position.

- High shoulder- the ladder is carried with the rungs vertical and the bottom beam resting on the member's shoulder. The arm under the carry will be used to stabilize the ladder by holding the top beam or a rung. This carry is specifically advantageous when throwing a ladder solo as it raises the breakover point allowing better angle of attack with the ground. The high clearance of this carry helps to avoid obstructions such as shrubs and A/C units etc. With repetitive practice this becomes most firefighters favored carry due to the functionality when throwing the ladder.
- Low shoulder- the ladder is carried with the rungs vertical and the top beam resting on the shoulder of the member. The arm will go through the rung gap and used to hold a rung stabilizing the rungs.
- Suitcase- the ladder is carried with an arm down at the side and hand holding the top beam with rungs vertical. This carry can be advantageous when overhead clearance is minimal i.e., traversing low slung trees.



Throws

The ability to throw a ladder without the use of a building or object other than clean ground is significantly advantageous as the opportunity to use the building is often not afforded to the member. Throwing ladders is a technique driven skill, more training will result in more effective positive results.

All ladders 24 feet (28 feet of Duo Safety) and shorter can be thrown by a single firefighter. Ladders greater than 24 will be thrown by 2 firefighters.

All PFD ladders will be initially thrown and used at a narrow angle then can be adjusted accordingly to facilitate rescues and egress for firefighters.

Always check for overhead obstructions on approach and prior to throwing the ladder. Powerlines can arch up to 10 feet to a grounded object. Have a plan for throwing the ladder prior to initiating the throw, extenuating circumstances often require a deviation from the initial plan. If the ladder begins to fall or the throw does not execute as planned, remain calm and gather yourself then work through the situation. Do not attempt to catch a ladder that is falling violently. Extension ladders thrown for uses other than rescue should have their halyard secured with a clove hitch or sufficient fire service knot.

- Beam raise- the beam raise is performed with the rungs vertical, planting the butt spur of bottom beam in the ground and traveling hand over hand until the beams are vertical. (Figure 2.)

- Flat raise- this raise is executed with both butt spurs planted in the ground and raised with the hands sliding the beams until the beams are vertical. (Figure 1.)

A properly thrown ladder should rest against the building at between a 60–70-degree angle, below the window, unless used for safety or as a secondary ladder.

Throw ladders with purpose on the fireground. If the ladder's purpose is horizontal ventilation throw it upwind on the wall adjacent to the window to be taken with the tip even with the top of the window. Ladders thrown for vertical ventilation should receive a second ladder creating a second means of egress.
If you are witness to a ladder rescue throw a second ladder next to the initial ladder, matching pitch, and length for personnel to use in assistance of the rescue.

Figure 1

Figure 2

Heeling

Ladders will ALWAYS be healed from the outside away from the structure. Healing the ladder away from the structure keeps the member away from falling debris or tools and gives more mechanical advantage to keep the ladder in place. When positioned away from the structure take advantage of this position and watch the work being performed above you. You are the second set of eyes for the member working on the ladder above you. **Ladders will be healed at all times** by at least a second member, unless secured in grass or tied to a structure. If two members are at base of ladder both will heal.


Rescue

Training regularly with portable ladders will have a great effect on the success of a rescue. When it is determined a victim needs to be rescued using a portable ladder, selection is paramount. Assess the height of the victim from the ground, obstructions on the building and the ground below the victim. Select the most appropriate length ladder to reach the victims location. A victim in a window will likely require the ladder placed with only the very tip inside the window to help avoid the ladder from washing out during transition. A victim on a roof will benefit from a ladder placed above the roofline at least 4 rungs to assist access to the ladder. If the ladder is selected and is determined ineffective in length, leave it in place and retrieve the correct ladder it may be needed later.

On approach select an area to execute your throw prior to starting the evolution. While approaching the victim, call out to them and attempt to calm them and coach them through the action that is about to take place. Keep watch on the victim while you are working. History has shown a victim will jump risking injury rather than burn in place. When throwing a ladder for an alert victim who appears excited and potentially irrational, throw the ladder out of their reach.

Once the ladder is prepared to reach the victim's elevation, move it in position via pivoting or rolling to avoid the victim grabbing the ladder tip and effecting your placement potentially causing the ladder to fall.

If a victim can scale the ladder independently allow them to do so, however, it may be beneficial to first ascend the ladder and assist them transitioning to the ladder and spot them while descending. If multiple victims are requiring rescue and the victim is confident and capable, allow them to self-extricate and assist the victim requiring assistance. Remember to interrogate victims and attempt to determine if others were at the location of the rescue. When assisting a victim remember to do a visual of physical search of the area near the ladder. Once the rescue is complete leave the ladder in place unless it is needed in another location for another rescue.

> Note: Ladders placed at a shallow angle can be less intimidating for conscious victims when possible. Steep angles can help with removal of unconscious victims, helping keep the body clear of the rungs.

If you are inside bringing a victim to a window for a ladder rescue constantly maintain control of the victim. Close yourself and the victim in an area of refuge if possible while awaiting the ladder. If the area is unable to be isolated from products of combustion bring the victim to the windowsill and control them from behind, attempt to control their limbs to prevent them interrupting the ladder operation or falling from the window.

Triage of ladders in rescue scenarios

Victims in immediate danger will receive rescue before all others on the fireground. Victims at or above the level of the fire will require rescue before victims below the level of the fire. Next selection will be based on conditions at or nearest the victim i.e., fire/smoke, same opening/next opening over. Remain observant during a rescue operation for multiple victims and constantly reevaluate their priority in rescue.

Care/ Maintenance/inspection

Ladders should be inspected at the beginning of every shift to ensure they are well maintained and operational ready.

Rig Check Inspection

- Roof hooks actuate appropriately in both directions and lock inboard securely.
- Butt spurs and tip rounds should be secure with all rivets and weld intact.
- Beams should appear straight and parallel with no cracking or heavy soot.
- Rungs should be tight with no rotational or side-to-side play.
- All rungs should be straight.
- Fly sections should fully extend with pawls clearing all rungs on extension.
- Ensure the pawls lock secure and equally on both ends of the rung.
- Ensure the pulley is free spinning and shows no signs of heavy wear.
- Inspect the halyard for imperfections.
- Secure the halyard to the bottom rung of the bed section only.
- Inspect for buildup of debris or soot on the entire ladder.

- Clean the ladder, if necessary, with a mild soap and bristle brush.
- Any finding that is concerning or causes you to be unsure should be IMMIDIATELY reported to the Driver Operator before continuing.

Report all findings to the Driver Operator for further decisions on repair or formal reporting of findings.

Weekly care

Apparatus day for ladders should include all the Rig Check Inspection as well as.

- Ensure all heat sensors are present and show no signs of exposure.
- Wash ladder with mild soap, bristle brush or towel. Steel wool may be used with consistent light contact for difficult debris.
- Inspect the ladder in sufficient light for any cracks or broken weld once clean. Run a dry towel over the ladder slowly feeling for snags caused imperfections in the ladder.
- Lubricate slides, contact surfaces and locking surfaces with paraffin wax only.
- Replace frayed or degraded halyard with supplied replacement.
- Dry the ladder before returning it to the stall on the truck.
- Paint 2 feet of the tip section and a 2-inch-wide stripe at the balance point (not necessarily the middle) with a highly visible color.

After fireground use inspection

After any ladder is used on an active fireground the ladder should be cleaned and inspected in a well-lit area. Check the entire ladder for any signs of direct flame contact as well as heat sensors for color change. Any ladder that shows signs of or visibly took direct flame impingement shall be taken out of service for further professional testing and inspection.

Section X: Search and Rescue

Overview

Rescue of victims in imminent risk of death or bodily harm is one of the prime objectives of not only truck companies, but all fire companies, and the basic reason for our existence as firefighters. All members **must** expect to put themselves in harm's way and take a significant personal risk to save lives. Effective training on search patterns, scene size ups, victim triage (determining who is in the most imminent danger), and victim removal techniques are required by all members. This will ensure victims are found and rescued in the quickest and most effective method, and to decrease the risk to the firefighters performing this fireground function.

The role of "Search and Rescue" typically is assigned as a Truck Company function, but due to many factors that are always changing and fluid on the fire ground, this may be assigned to any apparatus who arrives, and an immediate rescue is needed. Therefore, all firefighters regardless of apparatus assignment need to be familiar with this role.

Strategic Priorities

A first arriving truck company will typically split crews, 2 and 2. The two- firefighter crews will be designated High-Side and Low-side. (Refer to PFD Training Bulletin #21-11-Riding, Tool and Task Assignments). Depending on the location of fire and number of stories of the structure, both crews have the possibility to be assigned interior search. Therefore, all members of the truck company will need to perform their size-up prior to entering the structure.

Preparation for the search begins well before the alarm is received. It begins with the company level inspections and pre-plans, and EMS runs that the PFD performs on a regular basis. Also, being familiar with your districts that you cover, the types of construction of the homes located there and the layout of apartment complexes, to include the types of hazards, rescue problems, and accessibility issues will increase the chances of a successful outcome. The objective is to know in advance the approximate type and extent of search operations that might be required at any fire.

Size-up

Once arriving on the scene, the crew should continue with a thorough size-up, taking into consideration visual observations to establish the strategic priorities. The extent of fire, size and age of the building, population status in relation to time of day and type of occupancy are all important in ascertaining what rescue operations are needed. Other size-up information which will be pertinent is information obtained by fleeing occupants or who have escaped the fire building. Special urgency are reports of people who are still inside or unaccounted for, and this information should immediately be relayed via radio. On the other hand, reports that "everyone is out" might be erroneous. People in high stress environments and situations may inadvertently provide inaccurate information. Therefore, when assigned search of a structure, **<u>expect to find victims</u>**.

A quick but thorough size-up will also indicate where search operations should begin, the location of where victims will be in the most danger, and the likelihood of viable victims based on the conditions on arrival.

Direct Operations-

Direct Operations involving search are:

1. Removing victims from areas of immediate danger
2. Searching for and controlling fire extension.
3. Searching for and locating victims within the structure.
4. Actions that provide egress to victims and/or areas where victims are likely to be found.

Indirect Operations-

Indirect Operations involving search and rescue are:

1. Ventilation operations in support of rescue.
2. Isolating involved areas of the building, i.e., controlling openings and flow paths to keep the fire in check until rescue operations can be completed.

Tactical Consideration

Search Plan

1. Prioritizing the areas of the structure to be searched first.
    a. Victims in closest proximity to the fire room and directly above the fire need urgent rescue.
2. Access to those areas.
    a. Forcible Entry
    b. Points of entry, i.e., entering Charlie-Side vs Alpha-side.
    Or entering through a balcony door.
3. Consider VES.
    a. Stairwells to the upper floor of a structure may be compromised.
    b. Should be utilized when probability of finding a victim is high.
4. Designating an area of refuge.
    a. An area of refuge is the closest location that removes victims from harm, and where those victims are then handed off to treatment/ transport teams.
    b. The complexity of the problem will determine this location. The typical single-family dwelling, the area of refuge will more than likely be the front yard. Hotel or mid/high-rise fires, the area of refuge may be 2 floors below the fire, on the roof. A hand-off of victims in the evacuation stairwell may be appropriate as well.
    c. The area of refuge must always be manned.

Type of Occupancy

Apartments

1. The number of stories and distance from where an aerial apparatus that be placed and utilized can pose potential tactical and rescue problems.
2. Building design
    a. Does the apartment have a center hall construction or is it a “garden” type or central courtyard design with each unit having its own exit directly to the outside?
    b. If center hall or corridor design, search the fire apartment and apartments closest to the fire apartment first. If the hallway is “dirty”, or charged with smoke and visibility is severely limited or non-existent, search crews must perform an oriented primary search on hands and knees on the way to the fire apartment. Many victims in corridor style apartment buildings are found unconscious in the hallways and doorways after they had either attempted to escape in the smoke-filled hallway or opened their door to the apartment to investigate conditions and were met with smoke and gases.
    c. If the apartment is a “garden” style apartment building, then apartments that share the same exit landing should be checked, with immediate consideration to apartments above the fire, due to the ease of vertical fire spread from balcony to balcony.

Transient Housing

a. Transient housing, such as hotels, motels, etc., pose many of the same problems as apartment buildings. However, the potential for a high amount victims could be increased, due to the occupants of these structures will not be familiar with building layout or emergency exits.
b. Time of day for these structures will also pose significant rescue problems with the number of occupants utilizing the stairwells to escape. An evacuation and attack stairwell will need to be a high priority to be established very early on in the incident.

Schools

a. These have a high potential of victims who may or may not be able to help themselves, such as young school children.
b. Utilize the staff of these facilities whenever possible because of their knowledge of evacuation procedures, and headcounts of students and staff.

Commercial

a. Consider the size of the building and floor plan.
b. Victims in these structures are likely to become quickly disoriented because they are not familiar with the building.
c. Vast open areas may be present which increases the risk to firefighters. Consider search ropes.

Immediate Rescue

Immediate rescue must be attempted in extreme cases- such as when an arriving company finds occupants with immediate life threat visible upon arrival. Often is the case that these occupants will be located on the initial size-up and 360 of the structure. First arriving units will then need to perform rescue operations in favor of throwing ladders to begin rescue. This should immediately be reported via radio.

a. Get the attention of the victims and make it obvious that a rescue is underway.
b. Make every effort to calm the victims, talking to them until they can be brought down.
c. It is imperative that truck companies are prepared for this scenario and expect it. A plan must be in place with proper training and equipment for an efficient rescue of this complexity.
d. If you are presented with victims in this situation, it is likely that extreme heat and fire conditions exist in their room or apartment, and a ladder rescue will need to be made quickly and effectively, without error.
e. When throwing the ladder, throw the ladder outside of arms reach of the victim, as they will most likely try grabbing the ladder out of panic, which may complicate the situation further.
f. Once the ladder is thrown, walk the ladder over to the victim.
g. It stands to reason that there may be other victims trapped just inside the window. Strongly consider performing VES once the victim is removed to ensure no other victims remain inside that room.
h. Allow the victim to self-extricate if they are able to in order to decrease time and effort. However, there needs to be no doubt that the victim can safely do so without putting themselves or others into harms way by becoming more lost or disoriented. Example: The victim and rescuers can see the exit.

Ladder Rescue

(Refer to Ladders Portion of this Manual)

Search Procedures

Perform Initial Search Size-up

a. Layout
b. Square footage
c. Structural stability
d. Fire conditions
e. Search priority
f. Searchable areas

This initial size-up will determine your point of entry into the structure to begin search.

Go/ No-go?- Considering fire conditions, if we can safely occupy a space within the structure with your PPE, tools and your training, then the search is a **GO**.

Making Entry for Search

Once entry is made, do an immediate sweep for victims. This is performed by hooking a foot on the jamb, laying out full body length and sweeping from wall to wall. If the door is obstructed from opening all the way, reach around the door and check for victims behind the door.

Once you have checked for victims:

Check for ***<u>LIFE.</u>***

a. Call-out for victims, i.e., "Fire Department!" and momentarily listen for a response or coughing.
b. Scan with your eyes if there is enough visibility.
c. Scan with the TIC

Check for ***<u>FIRE</u>***

a. Look for a glow or crackling.
b. Which way is the smoke moving, how fast is it moving?
c. Utilize the TIC to check for heat pushing along the ceiling.

Note the ***<u>LAYOUT</u>***

a. Look for stairs, hallways, furniture.

<u>When making entry ahead of a hoseline being in place, search teams will make entry and control the door.</u>

Search Priority

a. Over 50% of victims of residential building fires are found in the bedroom. (USFA study of civilian fire fatalities in residential building 2017-2019)
b. Other common areas that victims are found are in the hallway and within 6ft of an exterior door.

Therefore, our search priorities are bedrooms and searching the egress areas as we move through.

Searching bunk beds, search the top bunk first, then the bottom bunk, then under the bed.

Victims in rooms behind closed doors have a higher chance of survival, so if we come across two rooms, one door is open and the other is shut, we search the open-door room first.

The fire apartment or room is the priority, with the hose team typically searching this room. In multi-floor dwellings or buildings, the adjacent rooms or apartments are the next priority, with another crew simultaneously searching the apartment or room above the fire room.

Truck Company Search

A truck company will split into two teams, High Side/ Low Side. A coordinated plan is imperative to decrease the risk of both teams searching the same areas on a primary search, and ensuring the teams are searching as much searchable space as possible in the shortest amount of time. Primary Search is fast, but thorough search carried out by the first arriving truck company. Primary search is when viable victims will be found and removed. Secondary search is a much more detailed search. A secondary is slow, and every area in that structure is searched. It is imperative for our truck companies to be proficient and effective at search and victim removal techniques. Victims must be found on the primary search, if they are not, their chances for survival are minimal at best.

Low-side will typically begin their search on the fire floor, nearest the fire or where victims are in the most danger. High-side will typically being a search on the second floor, above the fire or where victims will be in the most danger. On a single-story residence or structure, High-side will begin searching the opposite wall on a split-search or a split-targeted search.

When performing a search, it must be thorough and effective. Remember, we are expecting victims, and we are those victims' only chance of survival. **If you cannot see your boots while standing upright in a fire, you will crawl to give yourself the best possible visibility.** Drop to your hands and knees and perform a thorough search, using your hands to touch. Walking upright through a room or building will cause you to miss victims lying on the floor. Stepping on or kicking a victim while walking will feel like any other object and may not feel like a body. Searching with tools by extending the reach and swinging the tool out until we contact an object is not preferred. We could injure the victim further, and more importantly we will need to investigate the object we hit anyways to see if it is in fact a victim. There will be times were maintaining contact with the wall to stay oriented and providing a thorough search of all parts of that room will not be possible. Our job is inherently dangerous and will call on us to put ourselves in dangerous situations to save lives and rescue victims. It is imperative to get off the wall and search a room when needed. We do not need to maintain physical contact while searching for life inside a structure. Crews can still stay oriented utilizing different methods, like banging tools. Maintaining situational awareness, performing a constant size-up while moving through the structure by noticing conditions, locations of windows and doors are ways to maintain your orientation. Staying oriented is how we maintain the safest search that we can. This is not maintained by keeping physical contact with your search partner.

Searching for FIRE and LIFE

During our search, we are of course searching for life, for victims. Finding, "knocking down" the fire, if possible, and isolating the fire and fire room will significantly increase the possibility of a safer and successful search and rescue of a structure.

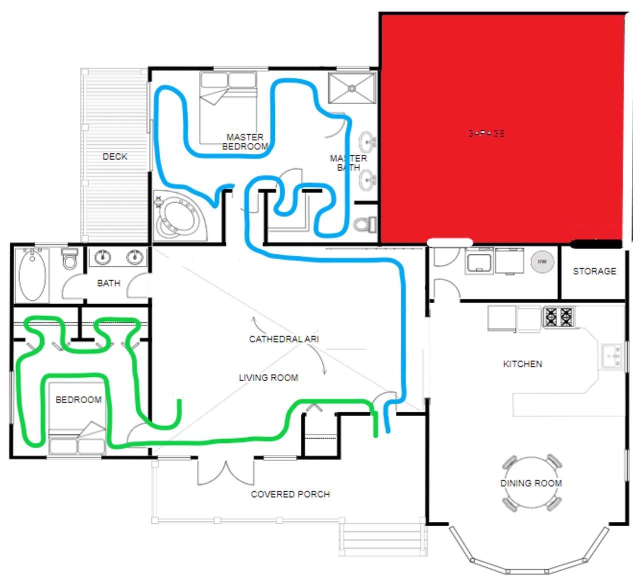
When we are searching ahead of a hose line and find the fire or fire room, we MUST communicate that immediately to fire attack. This will provide fire attack not only the location of the fire, but the best route to the fire.

When we find the fire and we have the water can, the search crew can make a quick attempt at knocking down the fire with the can. Whether successful or not, the next action will be to isolate the fire or fire room, if possible. Isolating the fire could include shutting the door to the room. If the door is burned off or non-existent, consider removing another door that will not cause an undesired effect on the fire, i.e., fire spread or dangerous flow paths, and place the door in the doorway to isolate the fire. Time is of the essence when searching for victims, and this should only be done when feasible. Once the fire is isolated, begin searching closest to the seat of the fire, working back from there. High-side should be performing a targeted search during this time.

Communicate and coordinate with other search crews and fire attack.

Verbally Communicate with other crews on what rooms have been searched. Using visual markings such as chalk or other means is not efficient, due to the probability of visibility being low or non-existent. Effective communication with other search crews will help ensure a rapid and effective search and help eliminate duplication of efforts by multiple companies.

Targeted Search

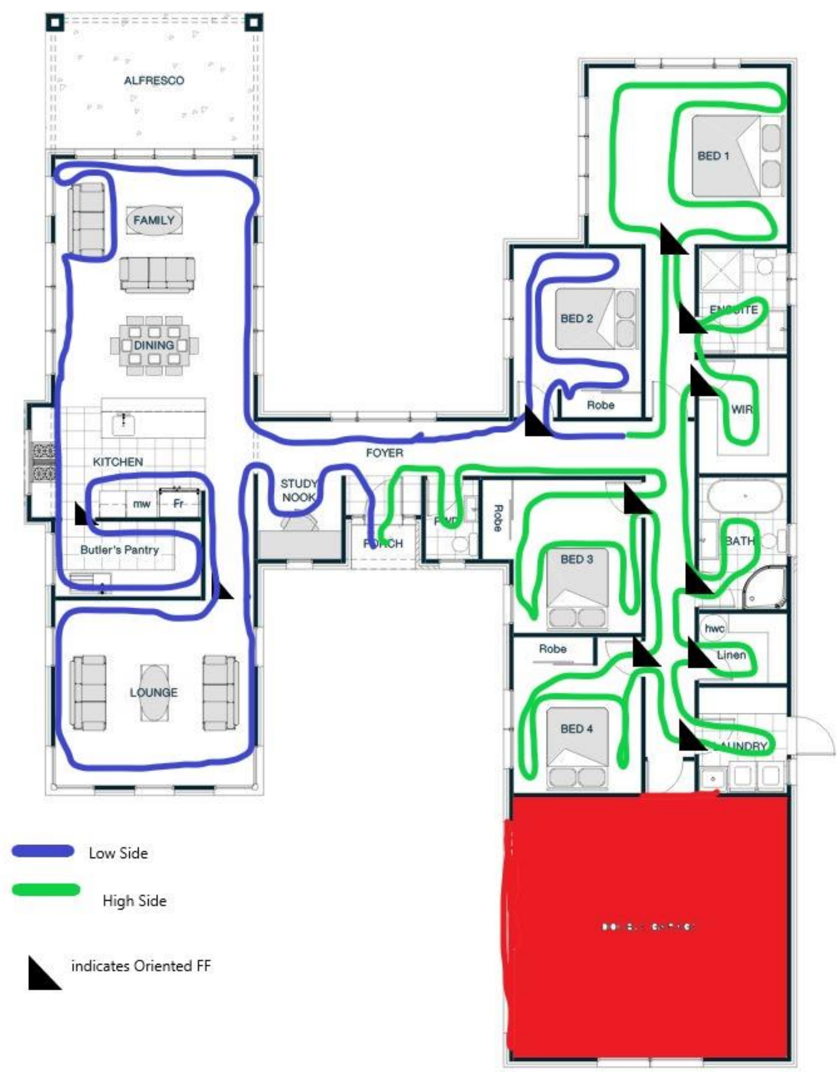


Targeted Search is performed based on what the exterior and interior size-up presents to that search company. Time of day is also going to affect whether the search crew is going to perform a split-search (right-hand, left-hand) or a targeted. At 0200hrs you as a company are assigned search, it stands to reason that a targeted search would be the best option. For example, when assigned search of a single-family dwelling at 0200hrs, most occupants will be sleeping. The search company should immediately enter the home and report to where the bedrooms are located, targeting the sleeping areas of the home. We as a crew on the fireground should have a good idea where the bedrooms are located, if we have performed an efficient and thorough exterior and interior size-up, noting the layout. If the fire room is located on the way to the targeted search area, crews should immediately isolate that room and report the location of the fire to the fire attack crew. Low-side and High-side crews will need to have a plan in place prior to entering the structure. If ingress to the targeted search area is cut-off by fire conditions, then VES must be considered and implemented.

If the fire occurs at 1400hrs on a weekend, the possibility of finding victims in the living/ family areas of the home is increased and may be more likely, and therefore may be targeted with priority over the bedrooms. Pay close attention to what the outside of the house is telling you on your arrival and 360. Are there children's toys in the front yard? How many cars?

Oriented Search

A traditional oriented search is typically performed when fire attack is not in place, there are poor conditions present, or an uncomfortable crew is performing the search. Oriented search is when one member of the crew is coordinating the search, typically the officer on the Low-side, and the DO on the High Side crew. All members should remain oriented, but the lead member is the one who is maintaining radio traffic, staying oriented to the progress of the search and the fire conditions. The oriented search also places a firefighter, typically the oriented member, at the threshold of a room that is being searched while another member searches the room. NOTE: While performing the oriented search, large rooms may not be able to be completely searched by staying on the wall, such as the family/ dining room and the lounge. Crews must be comfortable with getting off the wall to conduct a complete search of that room. If this is the case, then consider keeping the oriented member on the wall, and sending the search FF out into the open area, while maintaining verbal contact as well as contact with the TIC, or through banging tools. For large area searches, other avenues should be considered, to include search ropes.



Rescue Procedures

Rescue is the act of intervening and/ or removing a person from danger.

When we locate a victim, immediately advise that over the radio, and any needs you require, i.e., ladders, personnel, equipment. If you still have the capability to continue a search, if at all possible, try and hand off the victim to another crew. Once a victim has been found and broadcast over the air, there is more than likely another crew coming to assist. If this is the case, handing off the victim to them and continuing your search is easier to do than communicating what areas have already been searched, and will help maintain the orientation and integrity of the search. If this is not possible, or it will delay removing the victim from the IDLH atmosphere, then do what you need to do as a crew to get that victim quickly removed from the building. Remember, we are not performing a patient assessment, they are either obviously dead, or we are making the rescue. If we are making a rescue, then make it fast and effective.

Removing a victim

When removing a victim, there are several grips and techniques to utilize when doing so. It is not practical to rely on dragging victims with their clothing. Many times, during sleeping hours, victims will not be clothed or will have minimal clothing. The clothing they will be wearing will easily be pulled off or ripped, rendering them useless and not a viable option.

Dragging a victim foot first is preferred, in that it helps keep the victim's head out of the elements. These feet first drags can be performed by crossing the victims' legs, putting the victims' feet in each armpit, or using two firefighters to drag. The arm lock is another method to use if you find that dragging a victim headfirst is the better or only option. The time that this will occur is when we are dragging victims down a stairwell. This will always be performed headfirst as it will protect their head. Webbing, while it does have its advantages, can be cumbersome and time consuming. We have proven through training that this should be a last resort. In the time it takes to locate and remove the webbing and securely attach it to a patient with little to no dexterity due to your bunker gloves, the victim could already be well on their way to being removed. That said, having webbing in your gear for instances when obese or large victims are located could benefit the rescue more than hinder it. In our area, it would be extremely rare to encounter a basement or sub-level in a residence or business. However, should we encounter that situation, webbing would likely be the preferred option in moving that victim up the stairs and out of the basement.

Removing a victim out of a window

When removing a victim out of a window, place the patient under the windowsill, with the victim's toes touching the wall, knees bent. If a single firefighter is making the rescue, that firefighter will sit the victim in a sitting position, grab under the victim's arms and perform a wrist lock on the victim. The firefighter lifts the patient toward the window, placing the victim out the window headfirst, in preparation for rescue from the firefighter on the ladder or on the ground level. If two firefighters are performing rescue, the victim will be placed at the window the same way, but this time one firefighter will be on each side with the victim's arm. Securing the patient with an arm-lock, the firefighters lift the patient towards the window, and place him out headfirst for rescue.

Vent Enter Search

EXPECT VICTIMS, EVERYTIME.

All VES operations will be communicated to the fire ground and/or command.

Vent Enter Search places firefighters in an advantageous position within the building, through a window or other opening. VES can significantly increase the probability of locating a victim quickly and reduce the time it takes to make a rescue. VES is a targeted search of a bedroom or area of the structure that cannot be accessed by normal means due to accessibility issues, structural integrity, or fire/hazard conditions. Firefighters will also show access to a high risk area. VES can be performed when these conditions exist or there is a HIGH probability of victims being in that room.

LIFE is the number one priority on the fireground, and Pearland Fire Department will perform a search for trapped victims by any means if conditions allow.

> **Note: VES is not limited to the second or third division. VES can and will be performed in rooms on Division 1 when warranted.**

An example for a typical VES Scenario: Refer to the picture above. At 0300hrs, Pearland Fire Department arrives at this single-story, single-family dwelling and confirms a working fire. On the initial arriving officer's 360, the OIC reports bedrooms in the back of the home. Engine 8 stretches a line and makes entry into the front door and begins making a push. Tower 8 arrives on location and splits high and low side. Low side performs a 360 and performs a quick but thorough exterior size-up and notes the bedrooms in the back. Low-side advises that both High-side and Low-side will be performing a targeted search and will report directly to the bedrooms to begin searching for victims. Low-side enters the home through the front door and the OIC performs an interior size-up, LIFE-FIRE-LAYOUT. Low-side encounters the hose team from Engine 8 attempting to make a push but is cut-off getting to the bedrooms by fire conditions. No victims are seen in the immediate area. Low-side and High-side exit the structure and report to the back bedrooms to begin VENT ENTER SEARCH in the bedrooms. (It is important to remember that this will be situational dependent, as High-side may have remained outside to begin setting up for VES or horizontal ventilation. Also, the first arriving ambulance crew will be setting up for OV and will likely be the first crew to perform VES.)

Things to Note

If provided several different options, look for anything in the windows to indicate it is a bedroom, and begin your search there. Look for windows with a sash as most bedroom windows will have a sash. Large windows or plate glass windows usually indicate a game room, family room, or recreational room, and if located on Division 2, it could indicate the top of a vaulted ceiling. Children's toys in the windowsill, child curtains and stickers on the windows are also reliable indicators that it is an occupied bedroom. Bedroom windows per building code are to be windows that can open, generally having a sash in the middle. Windows that are non-opening are used for living room and other non-bedroom areas such as game rooms. If no windows show any obvious indications, quickly determine which room is the master bedroom. In general, the master bedroom is more typical in being used in homes with a single or couple living there. This size-up should be done within a few seconds. If there are no obvious signs of it being a master bedroom or an occupied bedroom, pick a window, preferably closest to the fire, and begin VES. Splitting a truck company crew into two crews and assigning both to primary search can be advantageous in this setting as well, as it allows simultaneous VES or multiple bedrooms.

Performing VES

Once determining which room to VES, the crew takes the window by striking the sash, knocking out all glass and "cleaning" the edges. The entirety of the window frame may pop out as well, which is favorable. The end of the ladder may be used to open a window, if a Firefighter is ready to quickly climb the ladder, apply their mask, and enter the room. Wait a few seconds to ensure any superheated gases and smoke do not ignite, since taking the window may have just created an exhaust point for the fire.

After waiting those few seconds, immediately sweep for victims that could possibly be found adjacent to the window. If no victims are located, and extreme fire conditions keep you from entering, back out, notify command, and proceed to the next window. If the room is searchable, sound the floor, then immediately enter. Remember, it is not up to us to determine survivable conditions, we only determine searchable conditions. If we can occupy an area with our PPE and our training, that area is searchable and will be searched. Victims come first. After sweeping for victims and sounding the floor, enter through the window or opening. Entering through the window with your head in the lowest position possible is preferred if hazardous conditions are present. Utilizing the “window-flip method” helps ensure this type of entry is made. Refer to the VES “Hands-on” Section of the manual.

** Use extreme caution when performing VES on any floor above Division 1, especially when searching directly above the fire room. Be sure to sound the floor to check structural integrity before entering the room, and constantly sound the floor during the entire search of that room.

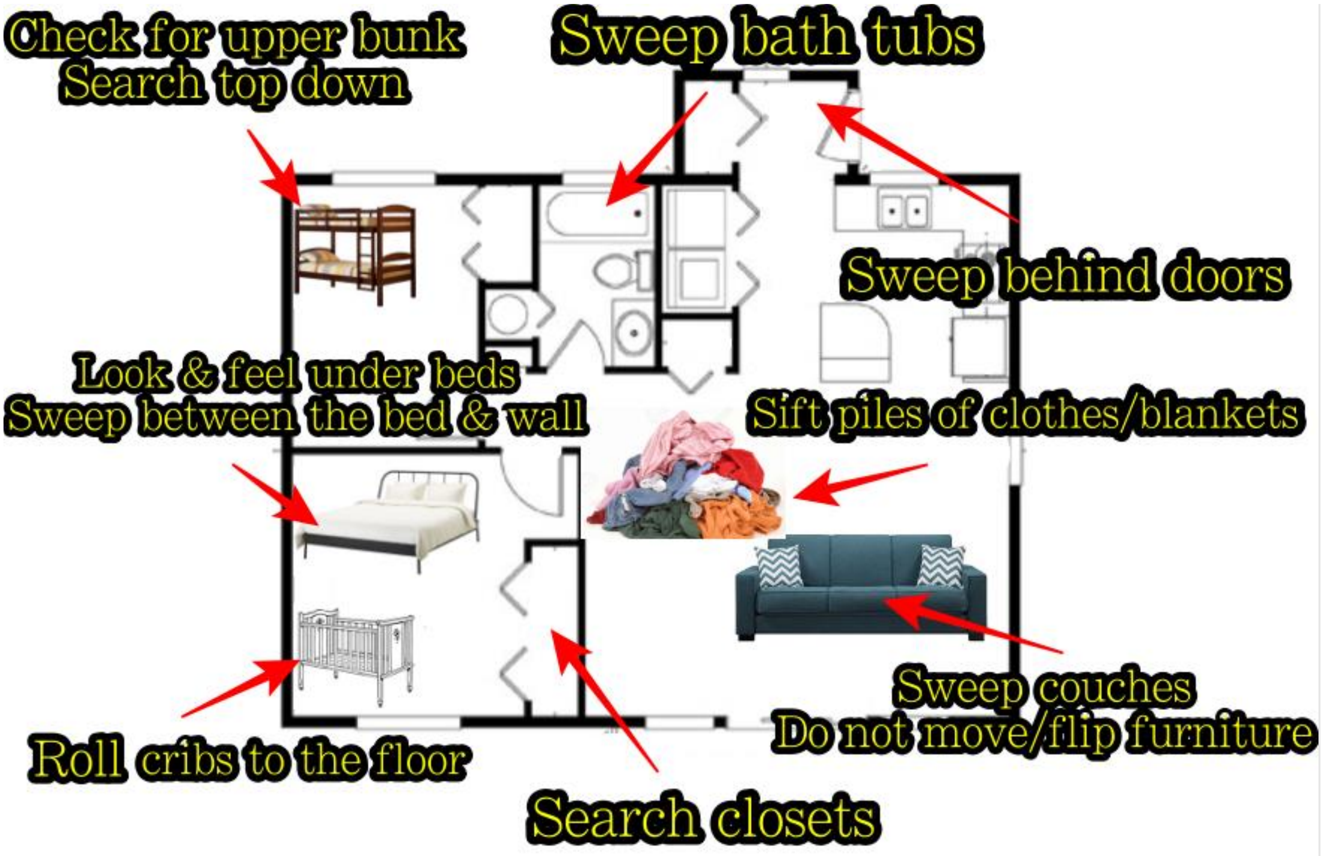
**Even if not operating directly above the fire, fire can easily spread between the floor, burning out structural members that can result in a collapse.

Once inside the room, the first crew member inside should immediately report to the door to isolate the room being searched. This is the one time that if a victim is encountered on the way to the door, you should pass that victim and get that room isolated. Isolating the room will keep a barrier between the fire and your victim, buy more time to report immediately back to that victim, remove the victim, and complete the search of that room. Note, before that member isolates the room, that member should perform an arm length sweep from the doorway into the hallway to check for victims. In the rare event that the door opens out into the hallway instead of inward, make sure you check behind the door for victims before isolating and returning to the room. As that member is isolating the room, the second crew member is at the top of the ladder, utilizing the TIC to help guide, providing constant communication and a reference point to the window for crew member 1. Depending on the circumstances, the second member can enter the room via the window. This can be for several reasons. If the first member finds a large victim on their search of the room, they may need help. The area may be larger than normal, requiring a two-person search. LIFE is the priority, two members splitting a bedroom for a search increases the chance of quickly finding and removing victims. Communicate the need for a third member to move to the top of the ladder if needed. However, with that said, the size of the room, or structural stability of the floor may dictate that a single FF entering the room will be efficient enough to perform a rapid search, and the crew will have to make that determination quickly.

If a large bathroom or closet is found larger than what a tool/ arm length search from the doorway can cover, crews are not to proceed further into that room. **VES is for the room you entered only.** Utilize a verbal callout and a scan with the TIC. If there is any verbal or audible indication that there is a victim located in rooms that extend further than the bedroom, only then may crews proceed into that area to save lives but must relay that via radio. Once the search is completed and no victims are found, crew members will exit. If one VES is being performed in one bedroom, then it is likely that other bedrooms will also need VES and could be carried out by the same crew. Rotating positions between crew members 1 and 2 can help keep the pace and intensity fast and high. Keep VES fast, but thorough. If victims are found, then utilize victim removal techniques found in this manual.

Search and Rescue Tips

- If visibility is low, then drop down low and search, that's where the victims are.
- Always be aware of where fire attack is operating, and the progress they have made on fire attack, and make sure they know where you are searching, especially when performing VES.
- When using a TIC, scan an area and then search. Try not to move forward while looking only at the screen, as tunnel vision can occur, and it provides poor depth perception.
- A room "clear" of victims utilizing only the TIC is NOT clear. Using a TIC is a great tool to assist, but a room needs to be cleared with a physical search. Heat signatures from victims can be hidden as victims are covered by blankets, clothes, debris, etc.
- When you locate a victim, check under and around that victim to make sure they were not carrying or assisting other trapped occupants.
- When taking a window, do not do so from directly in front, due to the potential for superheated gases to push out the opening and igniting.
- Doorways to the bedroom are typically located on the opposite wall.
- When encountering bunk beds, search the top bunk first, then bottom, then under. Then check behind the bed.
- Victims found in beds should be removed immediately, but crews should return to the bed and complete the search, to ensure no other victims are entangled or under the bedding.
- Do not move or toss around furniture. Doing so could distort your orientation, cover victims, or block egress.
- When isolating a room for a search, and the door is burned off, use a mattress to cover the doorway. This will buy you time to complete a search while isolated from deteriorating conditions.
- If you force an exterior door and a night latch chain is in place, this indicates there is a strong possibility that victims are inside.
- If you encounter a crib during a search, roll it to the ground to search. This keeps the victim and you low to the ground.
- Search piles of clothes, bedding, and curtains.
- If you do not have high visibility, drop down to the ground to search, that's where the victims will be located. It's impossible to thoroughly search a pile of clothes or behind beds by kicking and probing with your boots.
- When searching a bed, get as far onto the bed as you can, searching in a sweeping motion. Pull the sheets and bedding to you, until you feel resistance and investigate that resistance, it could be the weight of a victim.



Search Facts (Reference Firefighterrescuesurvey.com The first 2000 Rescues)

- **Were There Reports of a Victim v Total Recorded Rescues:**
  - Yes - Report of Victim(s) – 69% (1373/1996)
  - Yes – Report of Everyone is Out – 3% (61/1996) – *26% of these (16/61) times the report was given by a resident of the fire building.
  - No – No Report of Victim(s) – 28% (562/1996)
- **When Was Search Initiated v Total Recorded Rescues:**
  - Pre-Knockdown – 83% (1661/1991)
  - Post Knockdown – 17% (330/1991)
- **Did the Search Crew That Located the Victim Physically Have a Hoseline with Them v Total Recorded Rescues:**
  - Yes - 11% (72/670)
  - No – 89% (598/670)
- **Time From FD Arrival on Scene Until Arrival at Victim (Search Time) v Total Recorded Rescues:**
  - <2 minutes – 21% (408/1983)
  - 2-4 minutes – 30% (596/1983)
  - 4-6 minutes – 20% (399/1983)

- 6-8 minutes – 11% (219/1983)
- 8-10 minutes – 7% (137/1983)
- 10-20 minutes – 8% (150/1983)
- 20-30 minutes – 1% (21/1983)
- 30-60 minutes – 1% (18/1983)
- >60 minutes – 2% (35/1983

Vent Enter Search - The How To

Step 1:

Understand that putting victims' lives first is the reason for the fire service and firefighter's existence. We are their only chance at surviving these incidents, there is no one else coming. We must be prepared to put ourselves in dangerous situations to rescue people who are in danger. VES is one of those situations.

Step 2:

Quickly identify the room or rooms that will need VES. The occupants closest to the fire, either next to or above the fire, are in the most danger and those rooms should be considered first.

Step 3:

Communicate VES operations to the fireground and command.

Step 4:

Ensure correct ladder placement. The ladder should be thrown just under the windowsill. Placing the ladder inside the window frame will only hinder ingress and egress. Windows on Division 1 may also need a ladder.

Step 5:

Take the window. The best way to take a window is striking the sash with solid tool. Striking the sash will break the windowpanes, knock out the sash and often knock the window out of its frame, creating an entry point only after a few solid strikes. Be sure to note wind direction and speed and realize that taking the window creates an exhaust point the fire could work its way to that room quickly. When taking the window, do so from just under the window if on the ladder, or from the side if on the ground. If that room is filled with superheated gases and smoke, it could ignite. Directly in front of the window is not where you want to be. If using a ladder, extend the ladder to the finished length and place it parallel to the building. Use tip to break window by hitting sash in the center with beam of ladder. Immediately move the ladder into climbing position. Climb ladder finish cleaning out window. This needs to be done quickly as you are creating a flow path that needs to be controlled.

Step 6:

Clear the frame of any shards of glass. Reach in, keeping low in the window, and sweep inside the room, under the window, checking for victims.

Step 7:

Sound the floor to ensure structural stability. This is extremely important if performing VES directly above the fire room. (Figure 3).

Step 8:

Enter through the window or opening. Keep your head and upper body as low as possible, in the event the room lights off, you want to ensure you are not sitting up in it. Using the “wall-flip method” is the most desired technique. (Figure 4 & 5).

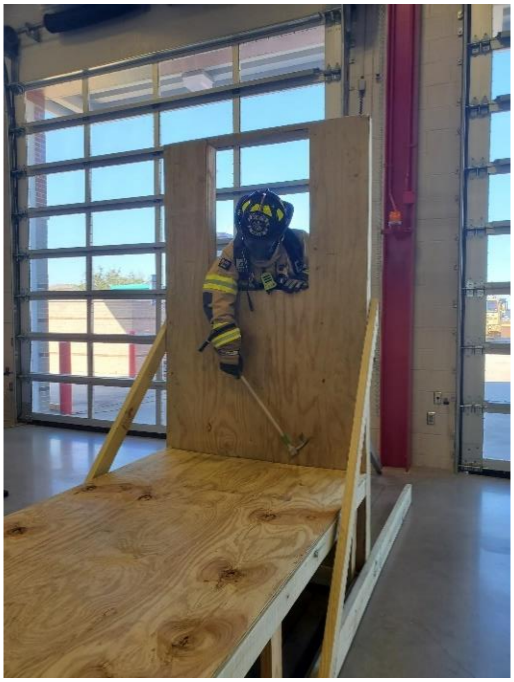


Figure 3

Figure 4

Figure 5

Step 9:

Proceed directly to the door to isolate the room. Perform a tool length search into the hallway before isolating.

Step 10:

Once the room is isolated, perform a search of the room. VES is quick, but it needs to also be thorough.

Step 11:

If a victim is found, call for extra hands, and begin removing the victim. Ensure radio communication. (Figure 6 & 7).

Figure 6

Figure 7

Step 12:

Conclude VES operations on radio to command.

Large Area/ Rope Search

(Coming soon to a truck book near you.)

Section XI: Ventilation

Overview:

Ventilation is an integral part of a **coordinated** fire attack and when applied properly can increase visibility, remove products of combustion, and heat from the compartment. This will offer relief for crews and victims inside the structure aiding the search, fire suppression, and victim survivability. Ventilation is a coordinated effort with fire suppression, if this is not done then ventilation can have a significantly negative effect. **Verbal coordination with interior crews must be utilized prior to and during any ventilation operations.**

Horizontal ventilation should be the predominant tactic when conducting ventilation. Horizontal ventilation is the preferred type of ventilation because it is simple, fast, and can be performed by crews carrying out their regular assignments on the fire ground.

In cases where horizontal ventilation is not feasible or effective vertical ventilation should be performed. Keep in mind vertical ventilation will only relieve the flow path from an exterior vent point to the vent hole. In other words, the vent hole acts and an exhaust, and intakes will need to be made for them to help products of combustion leave the structure. Remember to think about flow paths before making any openings.

Weather plays a vital role in ventilation. Personnel should make it common practice to be weather aware for the tour of duty, including a significant weather statement during rollcall is a good practice. Always ensure wind is working with you and not against you during ventilation operations, the results can be catastrophic and deadly. Remember wind in the forecast or at the Fire House might be different than wind at the incident scene. Ensure wind direction and condition on scene before ventilation is performed. Ventilation should never take precedence over victim rescue unless ventilation is what does victims the most good.

The use of fans will not be during fire attack and will only be used after the fire is under control and primary search is complete. Exception is to pressurize the evacuation stairwell in high-rise fire.

Horizontal ventilation

Horizontal ventilation is removal of products of combustion on the horizontal plane. This tactic can be achieved by opening windows, doors, and walls manually or forcefully. Initial horizontal ventilation should be performed opposite the attack ahead of the handline. Once the “fire under control” benchmark has been achieved the remainder of the structure can be opened manually by interior searching crews improving visibility and survivability. Any opening made forcefully breaking or removing intact compartment components will flow freely until the fire is extinguished, consider this before opening a structure.

Windows

Windows are often an excellent readily available option; however, they often require forceful opening from the outside. When taking windows on upper floors the window can be taken with the ladder or from a ladder. When taking a window with a ladder use the beam of the ladder to strike the middle of the sash to clear most of the window with one solid blow. Place the tip of the ladder beneath the sill. Climb the ladder and clear the rest of the window with a tool. To take a window from a ladder place the ladder tip at the same level as the top of the window to work on the upwind side.

Note: Ensure that window screens, curtains, and shades are removed as they can reduce ventilation through the window by 80%

- Residential windows should be taken by initially removing to the pane, then clearing the remainder of the window, starting at the top. Best performed with a roof hook.

- Plate glass windows should be taken with a blow to the top corner with a hook or striking tool. These windows will be more difficult as they are usually either storm rated or considerably thicker than common window glass.

Doors

Doors are often our primary access point into a structure and the first opening we create to affect the flow path.

Note: Every opening into a building we create will affect ventilation and should be considered.

When gaining access through a door it is best to leave the door operational so the flow path can be controlled.

Vertical ventilation

This method of ventilation offers a chimney effect to allow gases to travel in an upward direction. Time on the roof should be kept to a minimum as this is one of the most dangerous locations on the fire ground and it is difficult to determine the extent of roof member fire involvement. Vertical ventilation can be achieved on most roofs, however there are many difficulties due to modern construction practices and design. Newer lightweight construction weakens more rapidly under fire conditions versus legacy construction. The prior notion that a 4-foot by 4-foot hole is sufficient no longer stands true. Modern materials are releasing heat faster and in turn creating higher heat levels and more smoke. Hole size goals should be as large as necessary including the most amount of voids spaces and as close to the set of the fire as conditions allow. Some fires may require multiple holes. The crew must have a rehearsed plan

prior to deciding to vertically vent, including a cut method. When the crew is uncertain effective vertical ventilation can be achieved, another method should take priority.
Do not go to the roof to cut an ineffective hole! Cut the hole and get off! A proper sized hole is one that after the initial opening there is a dramatic increase in speed and volume of smoke leaving the hole.

Flat roof construction

While posing less fall risk flat roofs do pose fall threats such as walking off, stepping through translucent tiles, and collapsing. Flat roofs are commonly supported by truss systems or beams made of metal or wood, occasionally concrete. Flat roofs are commonly known to hold loads such as heating and cooling systems, communications equipment and various things businesses don't want seen at the floor level. Parapet walls are common surrounding multiple sides of the building to block the view of loading. Drain holes (scuppers) or down spouts will show the roof line offering a size-up of parapet height. Smoke conditions with poor visibility can cause significant hazard when piping and other raised objects are present. Sounding is paramount while operating on a flat roof due to the high likelihood of non-fire related hazards. These buildings can often be large and will require a "drive around" 360. Do not hesitate to continue around the building again to place an apparatus to support the operation. Always have a secondary means of egress when working on a roof (roof access, ground ladder, aerial ladder).

Flat roof material can vary from a multi layered layup with or without gravel to synthetic materials. Most commonly the base substrate of a flat roof will be either wood or metal and occasionally both.

Kerf cuts are an excellent means of confirmation of vent location. In the case you decide to abandon a location keep constant mind of the location of the kerf to avoid a fall hazard. If the kerf confirms the location of allow your cut to encompass the kerf.

Sky lights, translucent panels and roof hatches can be excellent means for quick vent when they are located near a cut location to offer some relief while the cut is made.

There are many different types of ventilators that can be located on a roof. Do not remove or damage any part of the vent. Rotating ventilators are 30% more efficient when operating as opposed to the turbine being removed. Ventilators are meant to ventilate, let them do their job.

Flat roof cuts

Opening a flat roof often reveals many surprising challenges due to the varying nature of construction. Bring a K970 equipped with a carbide toothed blade and chainsaw equipped with an RDR chain. These two saws will offer the best performance for most any construction you will encounter. If both saws are effective, consider using 2 saws simultaneously. If needed have a secondary crew there helping and offering a different perspective.

Trench cut

This process is a defensive effort in an offensive stance, writing off a portion of the building while actively working to save the remainder. Successful trench cuts incredible achievements as they are resource saturated and require high levels of coordination. The cutting crew will need to determine the location of the cut once on the roof. Once you have determined the location of the trench, open a small kerf to confirm the fire has not advanced to your location. Once you have confirmation there is no fire move rapidly towards the fire an open a large vent hole the closest you can safely work to the main body of the fire. This hole will stall the fire advance at this location allowing you precious time to complete the trench. The actual trench should run wall to wall and about three feet wide with handlines in place prepared to address the advance of the fire to the trench and stop it there.

Peaked roofs

These scenarios introduce additional fall and footing risk due to the pitch of the roof. Tactics may change with the pitch of the roof and access. A 360 size-up should be rapidly completed to determine the location of the seat of the and best location. Note the smoke behavior as related to the wind. Size up the roof pitch and composition prior to climbing when possible. Sound all roofs as if they have the potential to fail BECAUSE THEY DO despite the conditions below. Travel on high strength construction areas when possible. Make the largest cut you can at the highest point on the leeward side of the structure.

Peaked roofs with a pitch greater than 8 on 12 (8” of rise over a 12” run) will require a roof ladder to be used for travel to and from the work area.

Section XII: Outside Vent (OV) Position

Overview:

The outside vent position can be one of the most critical positions on the fire ground. The OV can make rescues, improve interior conditions, and create a more tenable environment for victims and interior crews. Because of this, it is crucial that the outside vent firefighter is well versed in fire behavior, building construction, reading smoke, understanding flow path, ventilation effects on interior fire, and be **HIGHLY DISCIPLINED.** The OV position IS NOT a freelance position. The OV is an assignment authorized by the department with specific tactical objectives, operating guidelines and strict communication and accountability standards.

Tool Selection for OV:

Our deployment model will dictate the minimum tool assignment for the OV crew. However, it may be necessary to grab additional tools. The OV assignment has assigned tools such as irons, 6' hook, water can, ladders, and thermal imager. Additional tools to consider *(for example on commercial fires)* consider an 8' hook, k-tool, rex tool, and saws.

The Role of the OV Firefighter:

The OV firefighter is assigned to operations mainly on the exterior of the structure (generally the bravo, charlie, delta sides). The OV position with PFD will consist of 2 firefighters whose roles are as follows:

- As the OV firefighters arrive on scene, they will conduct an immediate 360 exterior survey. Locating the fire and possible victims. A charlie side report/update can be given by OV as needed.
- OV will also shut off utilities to the building and radio crews and command when completed.
- OV will create victim/firefighter escape routes by softening the building and providing ingress and egress ladders to upper floors of the building. Radio progress/ladder locations to interior crews and command.
    - *Make sure to check inside and behind doors for victims. Use tools to extend reach into the building.*
- The OV will be ready to assist with horizontal ventilation coordinated with interior attack crews.
    - *Before making any opening for ventilation or rescue consider how openings will affect conditions. Make sure you account for wind direction, possible exposures, and where interior crews are working.*
- The OV position should be maintained throughout the duration of the scene if possible. Especially if no other officer has been assigned to charlie side as they will be the eyes for command on the back side of the building.

- *The OV should perform "hot laps" to continually monitor conditions. Track progress of interior crews from an exterior position. Vent as needed. Assist with removal of victims from exterior position.*

<u>If at any time the OV locates a potential for imminent rescue, they will radio command and conduct VES for rescue.</u>

MUST STAY IN ROOM OF ENTRY…NO EXCEPTIONS

<u>Outside Vent Crew Tips:</u>

Prior to taking windows remember:

When Ventilating remember fire travels the path of least resistance, up, across, down to oxygen supply. Breaking a window will intensify and draw the fire. Make sure you coordinate with search, hose and other vent crews.

USE EXTREME CAUTION WHEN venting windows remote from the fire room.

DO NOT TAKE WINDOWS between the fire and search/attack crew's point of entry.

DO NOT TAKE WINDOWS UNTIL READY. If the fire is in smoldering stage the introduction of air may cause a backdraft or rapid-fire extension.

Focus on taking windows in the fire room and the area around not involving search. Be aware of the status of water, attack, and search crews.

Coordinate with fire attack crews!

Reading Smoke Short Cuts (HINTS ONLY)

Black/Thick/Fast = heat and explosive

Black/Thin/Fast = flame near

White w/Speed = hot – but fire is distant

Uniform speed/color (steady flow & light color) from many places = deep seated fire

Brown = unfinished wood being heated

Turbulent = Flashover

Section XIII: Elevated Streams

Overview: Aerial Master Streams can be utilized for transitional attacks (I.E. Store Front Attacks, warehouse attacks) or defensive attacks (I.E. Elevated Streams).

All PFD aerial devices will be set up for offensive use and will have smooth bore staked tips in place and if equipped will have the ladder pinned to rescue mode.

Offensive Operations

Store Front Attack

The Bucket or Tip would be positioned in a way to flow quick and high gpm into a commercial business to get a quick knock down allowing engine companies to make an interior attack. The stream should be directed at the seat of the fire. THE AERIAL MASTER STREAM MUST BE SHUT DOWN PRIOR TO COMPANIES ENTERING THE STRUCTURE.

Warehouse Attack

Would be done in the same fashion but the tip/bucket would be placed at the opening deemed appropriate by the truck officer or truck chauffeur.

Defensive Operations:

Aerial Master streams would be placed in a way to place water most effectively on the fire. Keep in mind that roofs are meant to shed water and will inevitably prevent water from making it into the fire.

Ladders-The Tip of the ladder **will be unmanned** while flowing water.

Tower-The Bucket can be manned while flowing water.

Water Supply for Aerial Master Streams

The preferred method of water supply for ALL Aerial Devices will be an engine pumping the aerial device, not an engine supplying the aerial pump. Due to Friction loss in 5” hose at such GPM’s the engine must be no further than 100’ from the base of the Ladder/Tower truck. With 100’ of 5” hose, the pump pressure for:

- Ladders will be 160PSI for 1250GPM with Ladder at full elevation.
- Tower 180PSI for 1800GPM with Ladder at full elevation.

The Second method of water supply for aerial master streams will be Engine relay pumping into the Ladder/Tower on board pump with that pump then pumping the aerial master stream. In this case the engine and ladder companies must keep in mind friction loss between the Engine and Ladder. Our 5” is pressure tested at 200psi and we must keep below that on scene.

The Last method of water supply would be the Ladder is on a plug and becomes independent.

Ladder Truck Waterways

- Ensure that the waterway is pinned to the fly section rather than still in rescue mode where it is normally kept.
- Ladder Trucks will keep a smoothbore nozzle on the tip. If equipped with a fog attachment, it will be stowed in a compartment and attached when needed.
- Ladders/Tower while flowing water the turntable WILL be manned.

Section XVI: Truck Company Salvage and Overhaul

Salvage Overview:

Salvage efforts protect property and belongings from damage and will begin as soon as possible by the interior working companies. All companies should begin salvage efforts as soon as possible to preserve as many of the residents' possessions as possible.

Salvage Operations:

- Protection of victims' possessions
- Expelling smoke
- Removing heat
- Controlling water runoff
- Removing water from structure
    - Channel water to outside of building or outside drain
    - Water chute using ladders and salvage covers or with pike poles.
- Removing heat
    - Close doors after a room is searched.
    - Rapid ventilation (with coordination of fire attack team)
- Use of salvage covers (canvas or heavy plastic) over building contents.
    - Begin on floor immediately below fire room, move contents to center of room, cover with salvage covers.
    - Remove lighter valuables outside of building.
    - Use of special rolls and folding to be practiced by companies.
- Securing a building after a fire
- Covering building windows and doors or openings
- Patching ventilation openings

Safety Considerations during salvage:

- Full PPE required including SCBA.
- Beware of possible structural weakening.
    - Extra water weight
    - Heavy objects
    - Weakened structural components.

Salvage Tools:

- Salvage Covers
- Small Tool kit
- Vent Fans
- Squeegees

- Sprinkler shut off kit/wedges.
- Pike Poles to construct water chutes.

Overhaul Overview:

Overhaul is the process of searching for and extinguishing pockets of fire after the fire is under control. A fire is not fully extinguished until overhaul has been complete. Ideally fire investigators should examine the fire area before extensive overhaul begins to preserve possible evidence. Especially if arson is suspected.

Overhaul Operations:

- Identify and open any void spaces where fire might be burning undetected.
- Open any walls and ceilings to expose any burned areas.
- Be prepared for flare ups.
- Crews must be in full PPE including SCBA until environment can be ruled as non-IDLH.
- Consider using thermal imaging.
- Extinguish small pockets of fire using the least amount of water possible.
- Remove smoldering debris outside and douse with water.

Safety Considerations during Overhaul:

- When opening walls and ceilings be mindful not to block exits by debris.
- Utilize lighting.
- Limit the amount of water when and if possible.
- Utilize correct PPE.
- Look for signs of structural compromise.

Overhaul tools:

- Pike poles and hooks
- Prying tools
- Axes and saws
- Shovels
- Hooks and rakes

All crews on scene are responsible for salvage and overhaul. Our second most important job is preserving property. Overhaul prevents continued fire damage and salvage preserves belongings in the home. When tactically feasible and appropriate salvage and overhaul should be the top priority after search and extinguishment.

Section XV: Scene Lighting

Overview:

The ability for a firefighter to have improved visibility while operating at an emergency incident is incredibly beneficial to improve situational awareness and the ability for the firefighter to operate efficiently. Firefighters must be familiar with the location and operation of lighting equipment.

Lighting Equipment

- Fixed Lights
  - should be used to illuminate the emergency scene to improve overall scene safety.
  - When operating on an aerial the fixed lights on the tip of the ladder should be utilized to improve visibility when performing rescue, ventilation, etc.
  - When the area that needs to be illuminated cannot be done by fixed lights on the apparatus portable lights can be used.
- Corded lights
  - Can reach areas away from the truck but are limited by the length of the cord.
  - Continuous power supply, however, there is a possibility that a adapter is needed.
- Battery powered lights.
  - Lights can be quickly moved and placed in more restricted areas and are not hindered by cords.
  - Limited power supply.

Operations

- The Fixed lights on the apparatus will be the first line for lighting the immediate area of the scene.
- Lights should be used to indicate points of egress for interior crews as well as victims.
- When appropriate and feasible lights should be used to light immediate working areas of crews.

Section XVI: Aerial Ladder Maintenance and Inspection

Note: All the following are department minimum expectations for inspection and maintenance. Please refer to the truck's manual for additional maintenance needs and specifics.

Daily Inspection and Maintenance of the aerial ladder should include the following:

- Test the functionality of the outriggers, aerial ladder, lighting, and any other fixed systems for the aerial ladder.
- Inspect Ladder Section Sliding Surface
  - When inspecting, look for long scratches where the slide pads contact the aerial structure. Look for dirt, sand, debris, or metal shavings accumulating in the grease built up around the slide pads. If metal is present be sure to locate the source and correct any problem.
  - Inspect the ladder grease for excessive contamination and any dry or missing grease. It is necessary to maintain a film of clean grease on the sliding surface. If any time dry spots are found on the rails or any excessive contamination is found refer to the truck's manual and correct the issues.
- Inspect the hydraulic fluid level for the aerial ladder (this should include the basket leveling fluid if equipped). Add hydraulic fluid if found to be low. Refer to the truck's manual for the type of fluid and filling procedure.
- Visually inspect wire rope cables and pulleys to insure there are no frays in the cable or cracks in the pulleys.
- Check heat sensors are in appropriate locations and for any discoloration.
- If equipped: Ensure the breathing air system is full and operational.

Weekly Inspections and Maintenance (Apparatus Day)

- Complete your daily check along with the following.
- Inspect the Desiccant Moisture Indicators and replace them as necessary.
- Inspect Ladder Section Sliding Surface
  - Add grease to any dry spots as needed.
  - Note: For aluminum aerials Pierce recommends the ladder be lubricated every 3 hours of ladder operation or weekly whichever comes first.
  - Any time the grease appears to be contaminated (black or dark in color) consider degreasing and regreasing the ladder.
- Apply penetrating cable lubricant to the cables for the aerial ladder as needed. This can be done by wiping the cables with a rag saturated in the cable lubricant or by spraying the cable directly with the lubricant. Watch for overspray of dripping oil.
  - Note: These should never be degreased as is can cause rusting within the cable itself

- Apply lubricant to the waterway pipes of the aerial ladder and grease all waterway zerk ports as needed. Refer to the truck's manual for the type of lubricant to be used.

Quarterly Maintenance (Completed the first week of January, April, July, and October)

- Complete your daily check and weekly inspections and maintenance along with the following.
- Completely degrease and regrease ladder section siding surfaces.
- Apply penetrating cable lubricant to the cables for the aerial ladder.
    - Note: These should never be degreased as is can cause rusting within the cable itself
- Grease all waterway zerk ports. Refer to the truck's manual for the type of lubricant to be used.
- Completely degrease and regrease the stabilizer beams. Refer to the truck's manual for the type of lubricant and proper locations.
- Grease stabilizer pad swivel points. Usually, 2 zerks per pad. Refer to the truck's manual for the type of lubricant to be used.
- Grease Rotation gear by brushing on grease. Refer to the truck's manual for the type of lubricant and proper locations.
- Lubricate rotation bearing. Grease should be added while the aerial ladder is being rotated. Refer to the truck's manual for the type of lubricant and proper technique.
- Lubricate water monitors per the manufacture's recommendations.

Section XVII: Aerial Ladder Driving and Operation

General Statement: This operations guideline is just that, a guideline. All operators of an aerial apparatus understand that every fire scene is dynamic, and that they shall use their judgement on scene. Remember: **CITIZENS ARE THE FIRST PRIORITY IN OUR TACTICS!!**

Ladder Truck Driving Guidelines

The purpose of this section is to inform D/O's of the general items to be aware of when driving an Aerial Apparatus. This ***WILL NOT*** supersede PFD Procedure for Driver Safety, PFD Policy for Emergency Response, or PFD Policy for Backing. Please refer to those documents for anything outside of this manual.

Rear-Mount Straight Stick

- These apparatuses are not substantially heavier than our pumpers, however, the handling characteristics are quite different.
- Always scanning overhead obstructions while driving (i.e., tree limbs). These trucks are approximately 1-2 feet taller and will contact trees across the city.
- Being taller the center of gravity is higher and must be taken into consideration when making turns.
- The current single-axle Quints do not have a noticeable amount of tail swing difference.

Rear-Mount Aerial Platform (Tower Ladder)

- These apparatuses weigh approximately twice that of a pumper. Stopping distance **MUST** always be considered.
- These apparatuses are about 2 feet taller and same overhead obstructions and awareness is needed as in a Straight Stick.
- Rear-Mount Aerial Platforms have approximately 3 feet of overhang in the front due to the bucket, the bucket **WILL NOT** clear most street signs in the city.
- These are equipped with Tandem Rear Axles. The front axle is your drive axle. The rear axle will "scrub" (drag) when making turns, this is normal and will naturally cause a different tire wear pattern between axles.
- These do have a noticeable amount of tail swing when making turns. Watch mirrors when making turns and try not to turn when in tight spaces if possible.

All Aerial Apparatus

- The Aerial gets priority for placement on a fire scene. "Hose can be extended, ladders cannot."

- When making your final approach, start looking at building and fire conditions as well as man-made and natural obstacles. Place the apparatus in the spot that will give you the most use of your aerial for most of the building.
- Keep in mind that both types need approximately 18 feet to have the outriggers fully extended. If space is an issue **AND** you can effectively operate on only one side, you need a minimum of 13 feet to short jack the truck. (The Tower, being the widest truck when set, needs 12’ 9” to set one side fully extended and the other side set straight down.)
- Always think about placement and approach of your second due truck when spotting the first due truck. It may be a better option to only access part of the building and let the second due truck access the other part for more coverage.
- Take into consideration the 75’ aerials when placing the 100’ aerials. The 75s may need to be in a closer position since the 100s have the extra reach.

Ladder Truck Operations Guideline

Driver Responsibilities

- Tour check of apparatus including operation of truck and status of aerial device
    - Aerial Device Master Stream nozzles must be stowed with the nozzle facing forward and out away from the windshield to prevent collision with windshield when stowing the aerial device.
        - Additionally, no stream-straighteners should be placed on aerial master stream devices without approval from Operations.
- Assure all equipment is operational for set deployment model.
- Assure that truck is set up for high side operations, and ready to split crews.
- Assure accountability passports are accurate for crew.

Aerial Operations on Multi-Family Residential Occupancies

- Perform drive around 360 if possible.
- Spot apparatus on corner of the building involved for greatest scrub capabilities.
- Driver should, as time allows, have outriggers set and aerial device positioned for rescue opportunities/ means of egress.
- Should D/O be performing high-side search/roof duties, the next arriving High Side should take over operation of aerial device.

Aerial Operations on Single-Family Residential Occupancies

- D/O shall place the aerial at the most advantageous location, preferably the front of the residence; to allow access to windows, for second floor, and roof access.

- Arrival on scene, D/O should, as time allows, have outriggers set, and get aerial ladder un-bedded and to a window if not in defensive operations. If a single-story residence, aerial should be placed at roof level for vertical ventilation.
- Master stream device should always be pinned to “rescue mode”, moved only when needed for defensive fire operations.
- If using aerial devices for rescues or rooftop ventilation, there should be an approved operator on the turntable to allow for movement of the aerial device if needed.

Aerial Operations on Commercial Occupancies

- D/O shall utilize apparatus if able to perform a “drive around” 360 of the structure to identify areas to access rooftop.
- If needed D/O should position apparatus to allow for a storefront blitz attack. Otherwise, it should be positioned in the most favorable position for roof access and a roof report.
- Operators should position the apparatus on the corner of the structure, to allow for some protection for collapse if possible.
- If applicable the master stream device should always be pinned to “rescue mode”, moved only when needed for defensive fire operations.
- Positioning for straight stick aerials should be advantageous for allowing access and egress for firefighters. Positioning tower aerial apparatus should be advantageous for rescuing victims, and defensive operations.

Aerial Operations for rope rescue

- D/O and Officer shall locate the area best suited to perform rescue using aerial device.
- <u>Note:</u> Working directly off the back of the truck is considered one of the strongest areas to work.
- Aerials should be pinned in rescue mode to allow for full use of the aerial without obstacles in the way. Tower operations should be the same without pinning necessary.
- When possible, the patient should be packaged and secured within the tip of the aerial ladder and moved to the ground accompanied by a firefighter. This would not use rope systems and is the simplest/ safest way to move individuals.
- Should rope systems be needed rescuers should attempt to secure the patient to the tip of the ladder without the use of MA (Mechanical Advantage) systems and boomed to the ground.
    - <u>Note:</u> This should only be done when there is no risk of the load snagging on nearby objects, and the load can be safely monitored by rescuers. **If there is a potential of the load getting caught or snagged on objects this method should not be used.** When using the aerial ladder there is no way to know

what forces could be placed on the load should it get caught or snagged by objects nearby, so low speed, and extra caution should be used. A rescuer should accompany the patient whenever possible.

- When needed the truck can be used as a high point anchor.
    - Note: It is important to remember that ladder trucks are considered weak anchors and you need to understand the forces that are being applied by the MA system. Often MA systems are doubling the load if not more. For this reason, it is recommended that the load be kept to a single person load and that there are no individuals on the ladder.
    - When utilizing aerial as high point/change of direction, the apparatus should be set, then turned off.
    - D/O should perform walk arounds to make sure any ropes being utilized are not being compromised by sharp edges, or exhaust from the truck running prior to rescue being performed.
- Spotter(s) should be assigned to monitor aerial tip, truck, and load.

Other Items

- "Kissing" the structure is acceptable; if no significant pressure is being exerted onto the aerial device, and the truck is backed off of the building.
- If "kissing" the structure to affect a rescue; the operator on the turntable and the rescuing truckman shall remain in constant communication and operators hand off the turntable controls until given the direction by truckman to move.
- D/O should approach victims in windows or elevated surfaces from above or the side rather than below to prevent fire victims from jumping onto the aerial device, unless approaching from above would put the victim in a more compromising situation.

Water Application

- D/O should verify master stream device is pinned in stream mode.
- Spotter should be used on ground to verify that other members are clear of water being applied.
- Constant communication should be in place between turntable and firefighter if a firefighter is used on the aerial while water is being applied.
- An abundance of caution should be practiced if moving the aerial while flowing water, (refer to operator's manual to review limitations)
- At no time will a member be on an aerial ladder flowing water. (The only exception will be the tower truck)

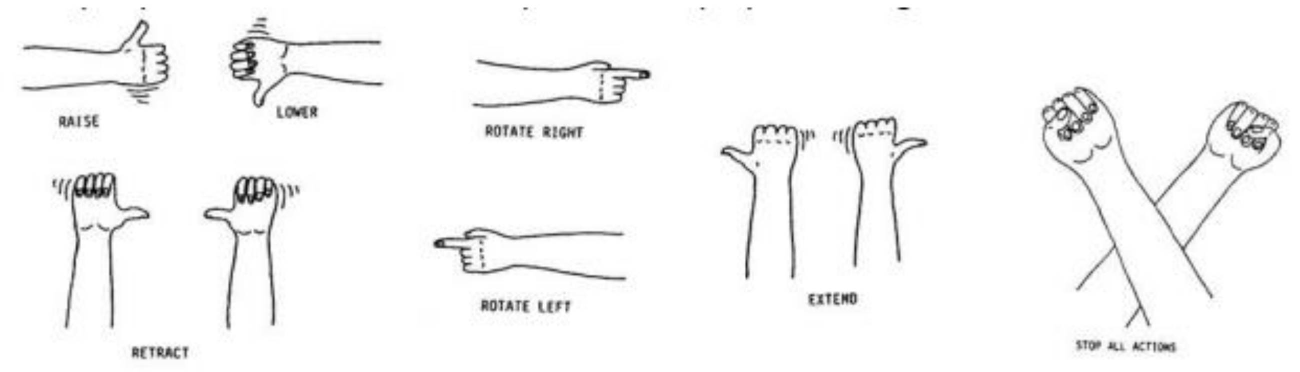
Hand signals

Often the aerial operator cannot see accurately spot the aerial device due to the position at the pedestal. It is often beneficial to have a spotter standing at a better vantage point to help guide the aerial operator.

Procedures for hand signals:

The spotter should stand in a position where he/she can observe the aerial device and any obstructions that may encounter the aerial. They should also be in a position where the aerial operator can see them and any hand signals they might be giving. The spotter will use the signals described below to guide the aerial operator into the objective.

Normal Visibility:



Low Visibility/Night Ops:

Use of flashlight will be required and will use same signals as normal with exception of “stop”. This will entail moving flashlight rapidly from side to side